



SOICT

HUST

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HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

Multimodal Machine Learning

Lecture 4.1: Multimodal alignment

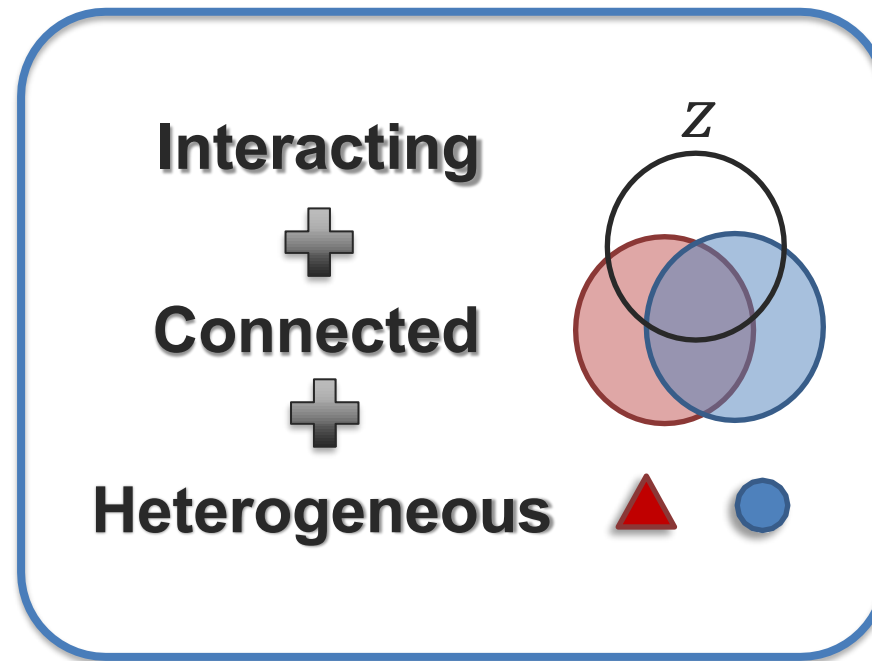
Dam Quang Tuan

Lecture objectives

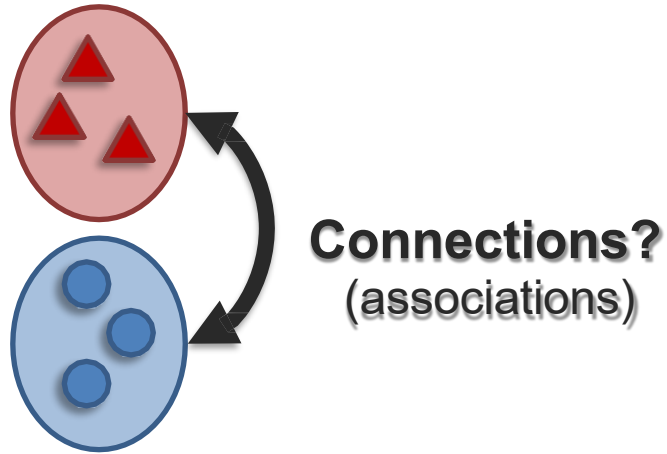
- A quick review
 - Connections, coordinated representations and mutual information
 - Modality interactions and factorized representations
- Discrete alignment
 - Local alignment
 - Coordinated representations; hard and soft attention
 - Global alignment
 - Assignment problem and optimal transport
- Continuous alignment
 - Continuous warping
 - Dynamic time warping
 - Discretization and segmentation

A Quick Review

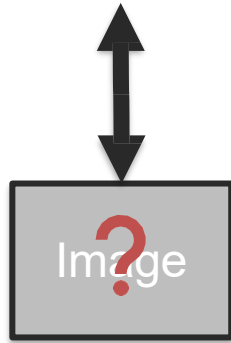
Multimodal is the scientific study of
heterogeneous, connected and interacting data



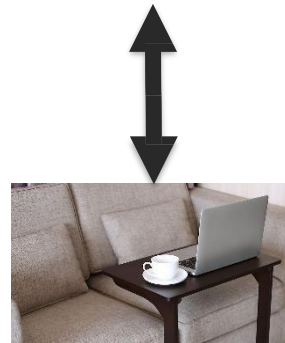
Connections between Modalities



"kitchen"



" ? "



Connection types:

●—● Co-occurrence

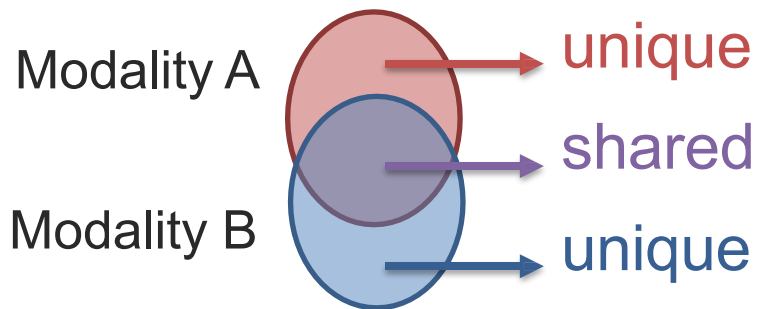
●—● Correlation

●—● Causality

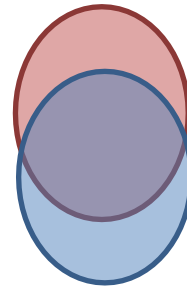
●—● Relationship

⋮

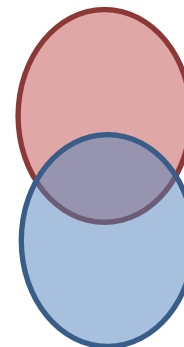
➡ knowledge of one modality provides information about the other modality



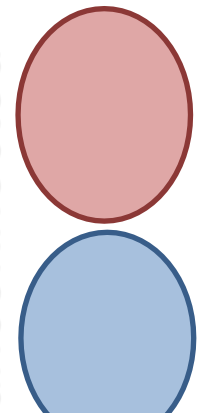
Strongly
Connected



Weakly
Connected

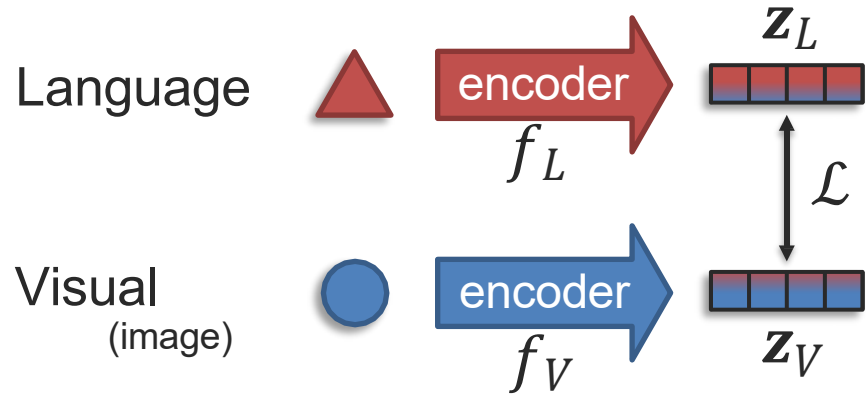


unconnected

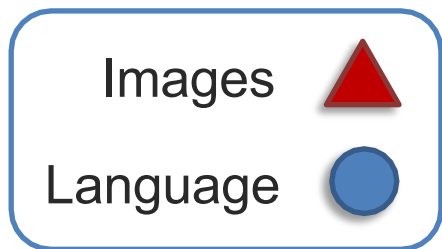


Connections are part of the data... and models will try to learn them.

Coordinated Representations – Example: CLIP



Connected modalities:



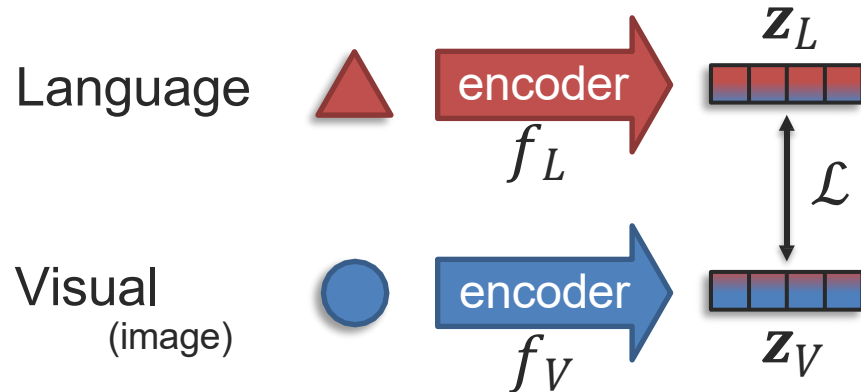
Popular contrastive loss: InfoNCE

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\text{sim}(\mathbf{z}_A^i, \mathbf{z}_B^i)}{\sum_{j=1}^N \text{sim}(\mathbf{z}_A^i, \mathbf{z}_B^j)}$$

Annotations for the InfoNCE loss formula:

- $\text{sim}(\mathbf{z}_A^i, \mathbf{z}_B^i)$ (green box): positive pairs
- $\sum_{j=1}^N \text{sim}(\mathbf{z}_A^i, \mathbf{z}_B^j)$ (red box): negative pairs and positive pairs
- Similarity function can be cosine similarity (orange text)

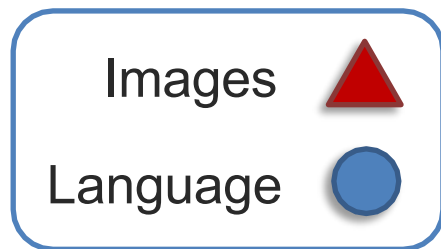
Coordinated Representations – Example: CLIP



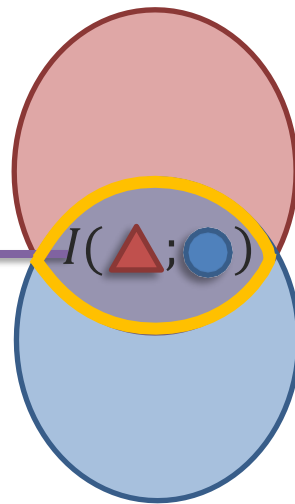
Popular contrastive loss: InfoNCE

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Connected modalities:



shared



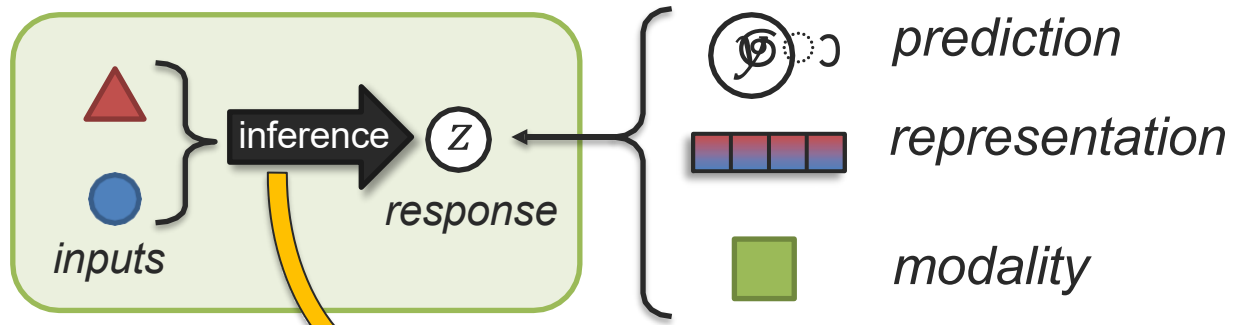
Mutual information $I(X; Y)$

$$\mathbb{E}_{X,Y} \left[\log \frac{P_{XY}(x, y)}{P_X(x)P_Y(y)} \right]$$



CLIP focuses on shared connections

Modality Interactions

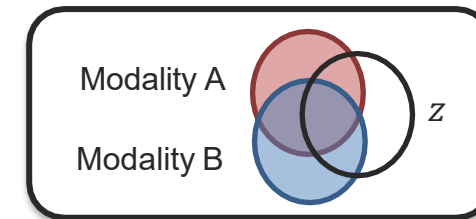


**Interactions happen
during inference!
(from human or model)**

Interactions require more
than the input modalities!

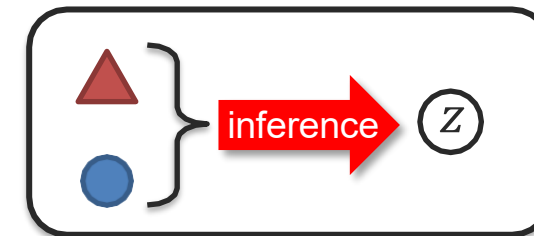
Interactions taxonomy:

Level 1: Response(s) and Input Modalities



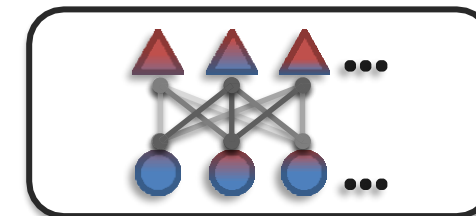
- ☐ Co-occurrence
- ☐ Redundancy
- ☐ Dominance
- ☐ Emergence
- ...

Level 2: Interactions – Internal Mechanics



- ☐ Additive
- ☐ Multiplicative
- ☐ Polynomial
- ☐ Nonlinear
- ...

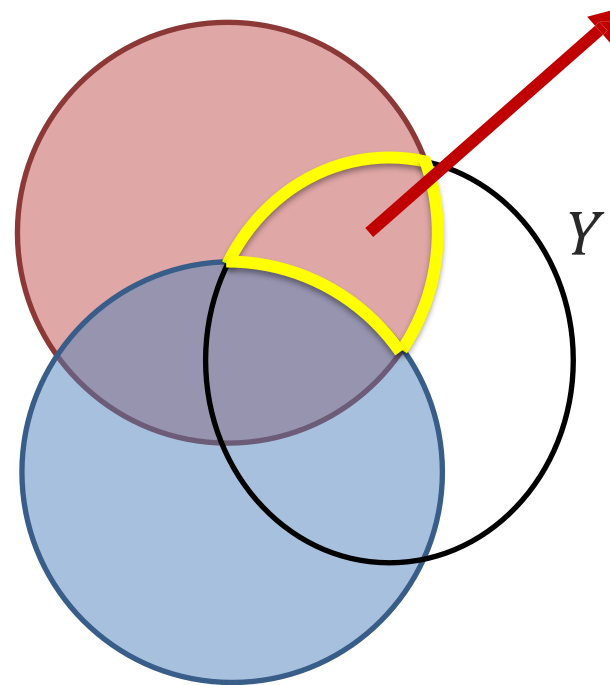
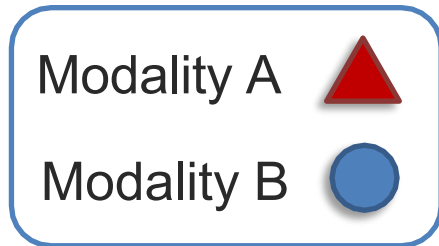
Level 3: Contextualized Interactions



- ☐ Temporal
- ☐ Hierarchy
- ☐ Multimodal
- ...

Modeling Interactions – Response and Input Modalities (Level 1)

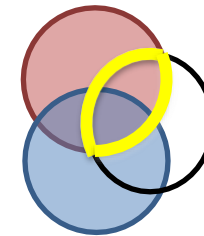
Information theory as a framework for modeling Level 1 interactions between input modalities and responses



Task-relevant **unique** information

$$I(\triangle; Y | \bullet)$$

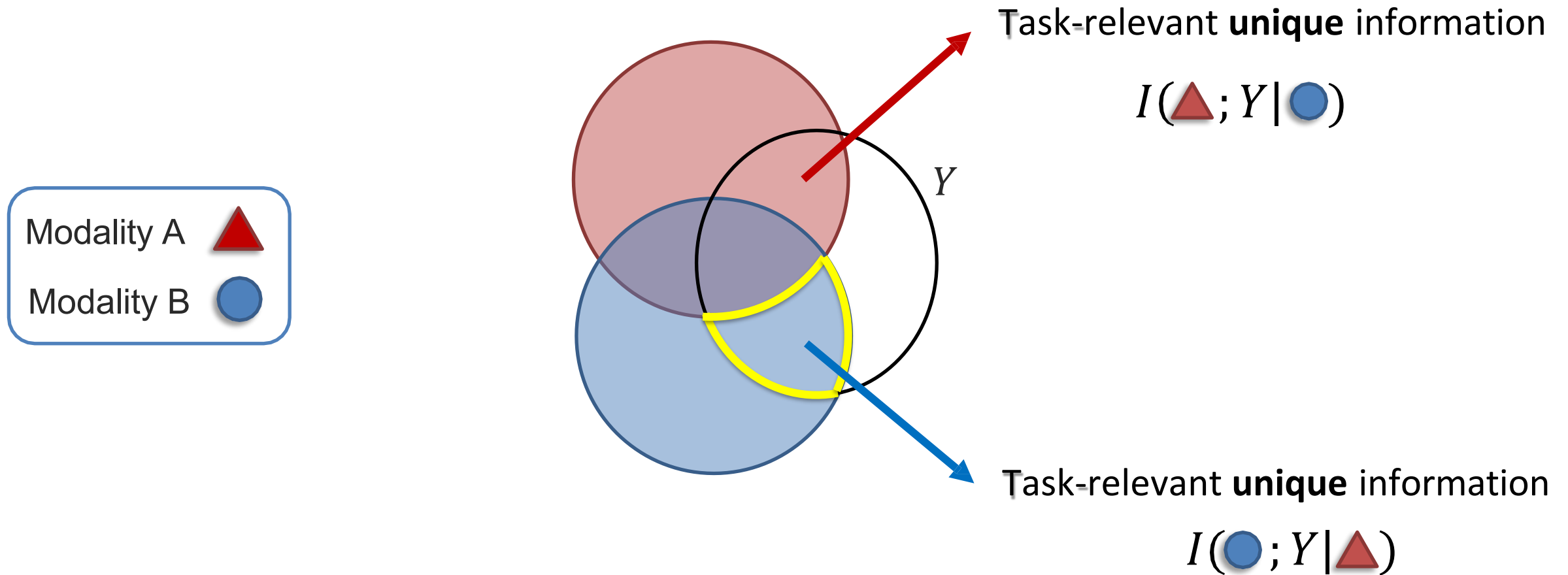
Mutual information...



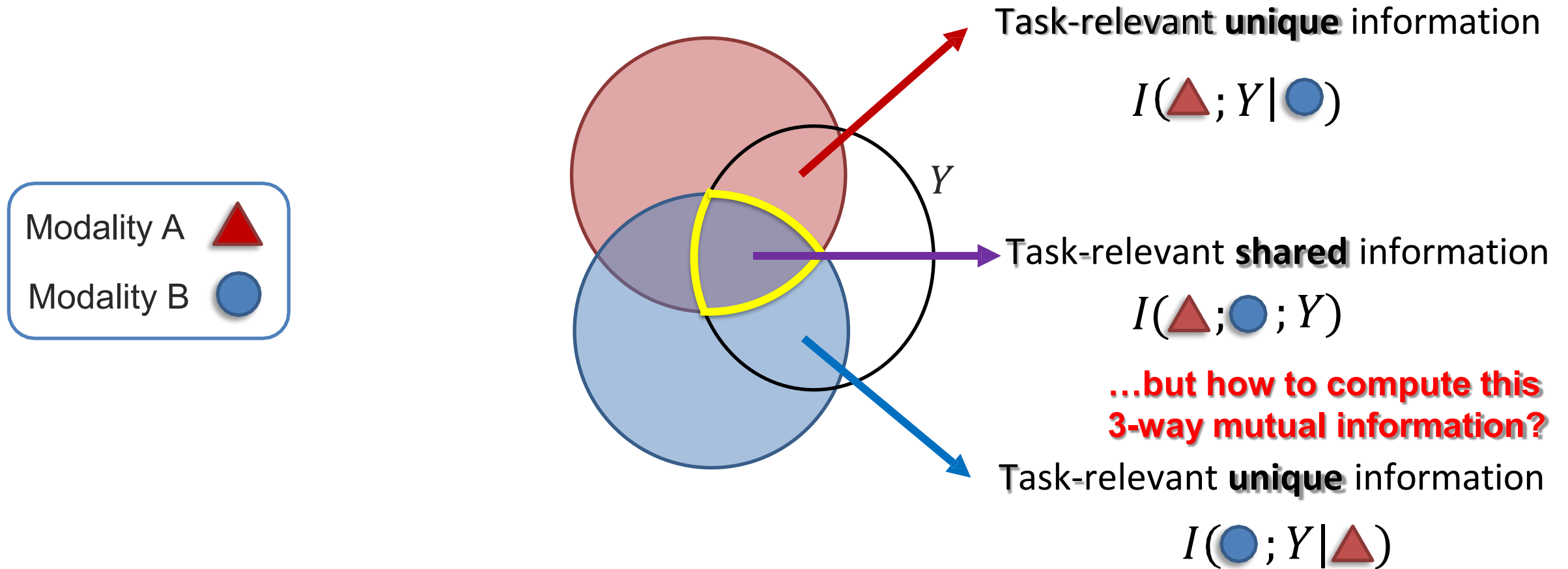
$$I(\triangle; Y)$$

...but without 

Modeling Interactions – Response and Input Modalities (Level 1)



Modeling Interactions – Response and Input Modalities (Level 1)



Modeling Interactions – Response and Input Modalities (Level 1)

Partial Information Decomposition (PID)

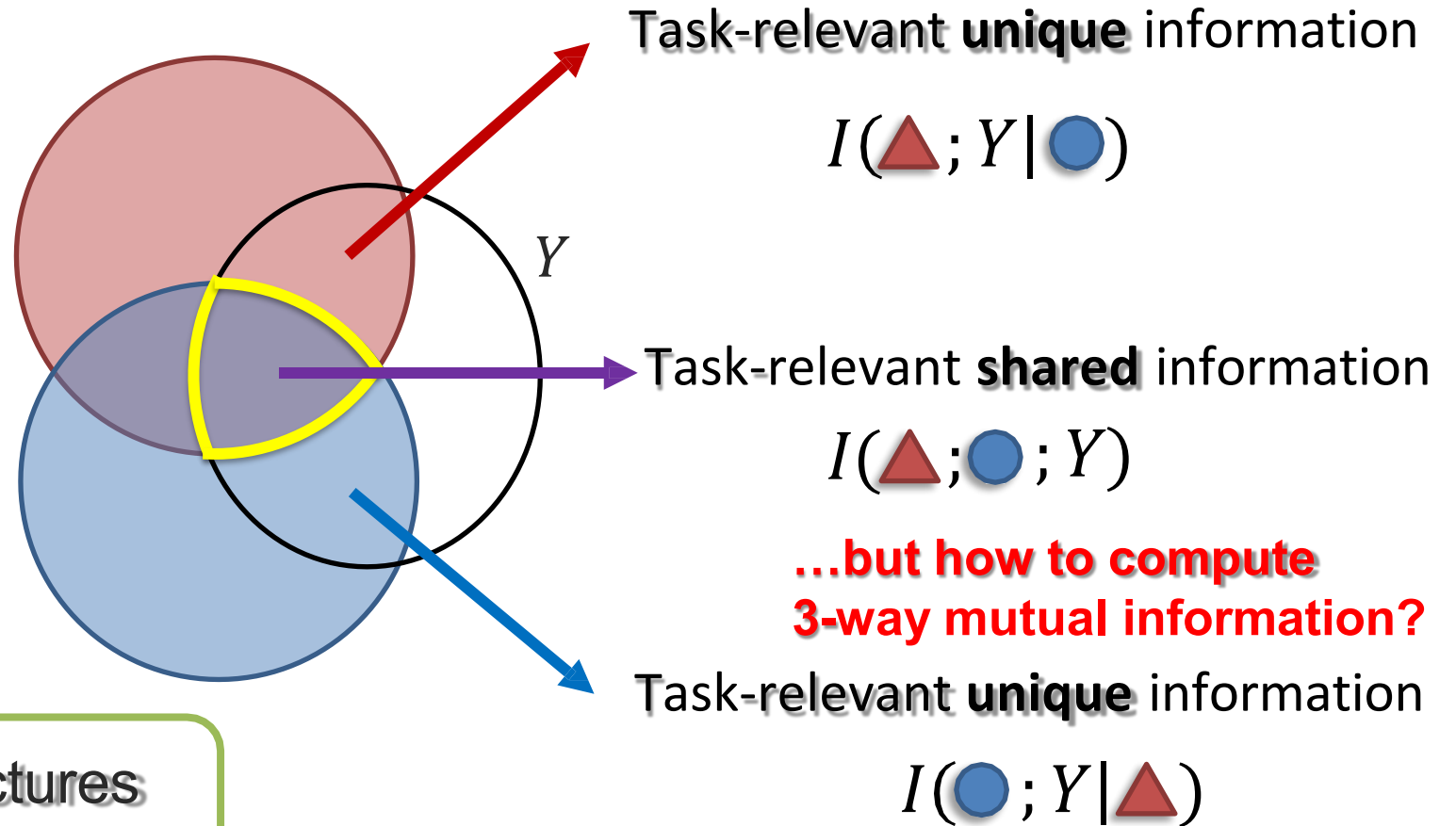
$$I(\triangle; \bullet; Y) = R - S$$

factorizes 3-way mutual information into:

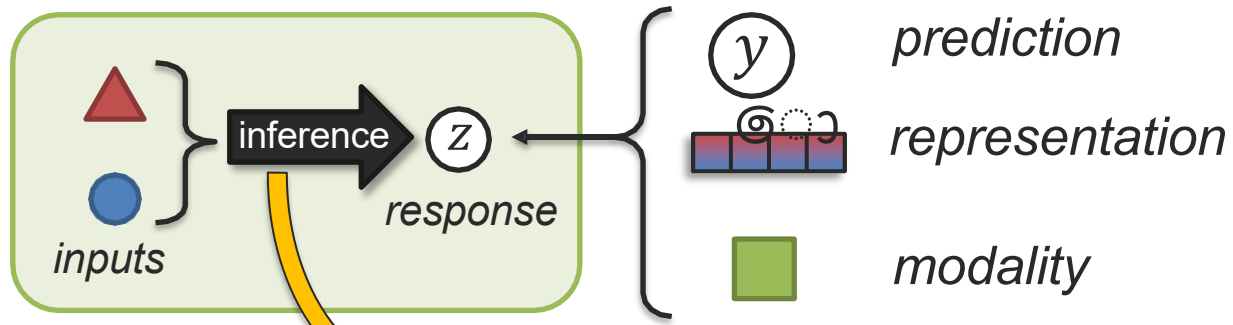
R: redundancy

S: Synergy

More about PID in future lectures
and reading assignments



Modality Interactions

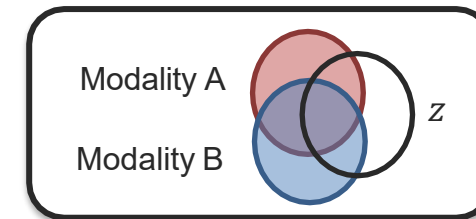


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Interactions require more
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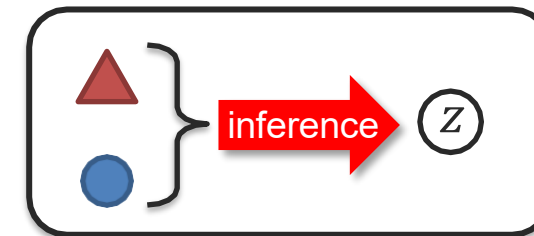
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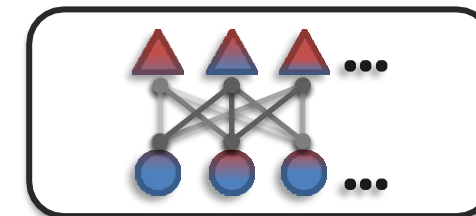
- ☐ Co-occurrence
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- ...

Level 2: Interactions – Internal Mechanics



- ☐ Additive
- ☐ Multiplicative
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- ...

Level 3: Contextualized Interactions



- ☐ Temporal
- ☐ Hierarchy
- ☐ Multimodal
- ...

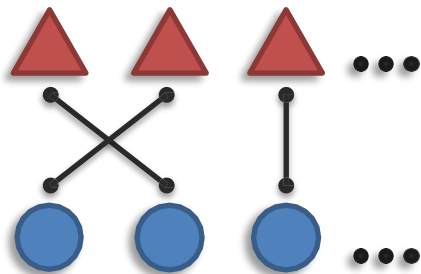
Challenge 2: Alignment

Challenge 2: Alignment

Definition: Identifying and modeling cross-modal connections between all elements of multiple modalities, building from the data structure

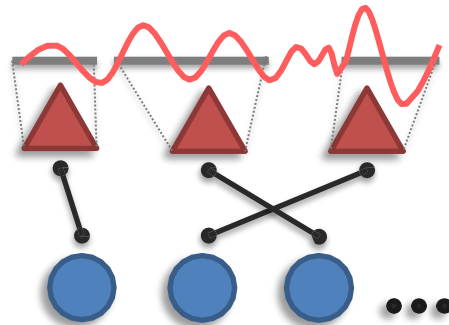
Sub-challenges:

Discrete Alignment



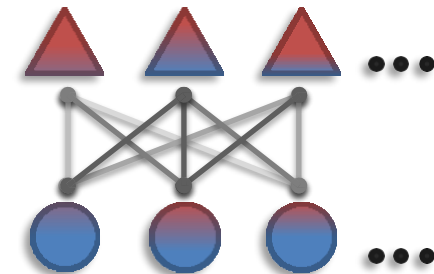
Discrete elements
and connections

Continuous Alignment



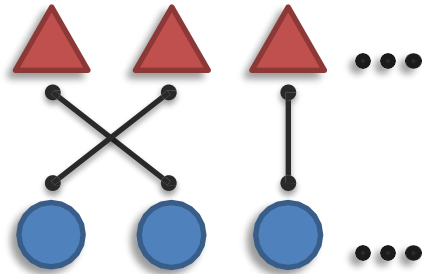
Segmentation and
continuous warping

Contextualized Representation

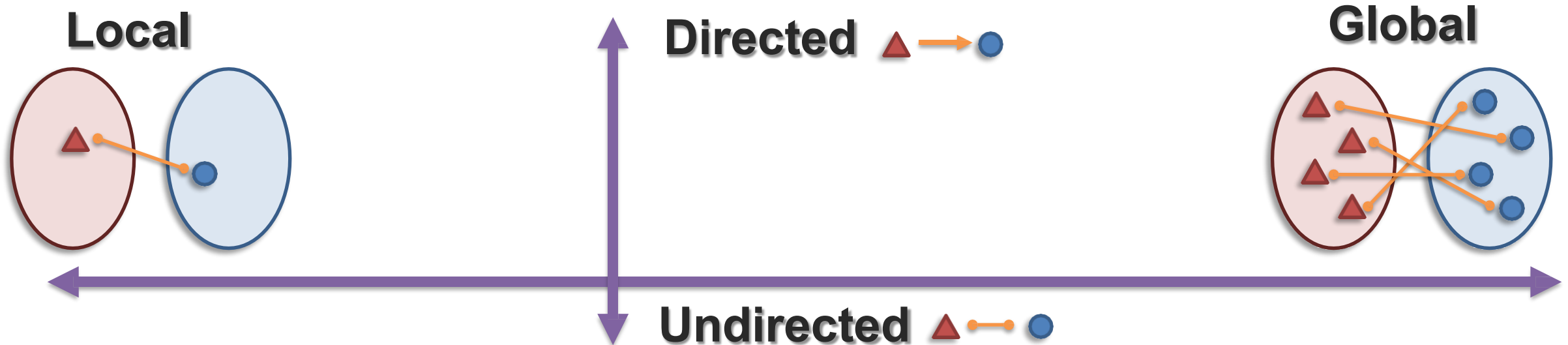


Alignment + representation

Sub-Challenge 2a: Discrete Alignment



Definition: Identify and model discrete connections between elements of multiple modalities



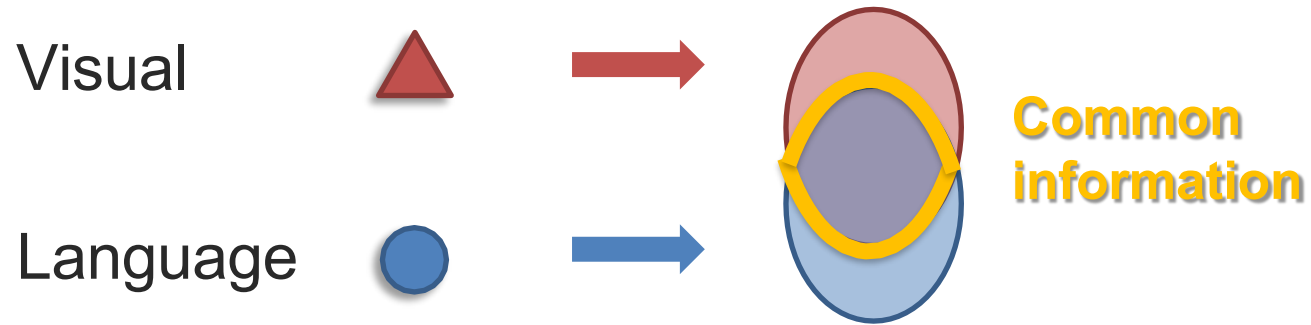
Language Grounding

Definition: Tying language (words, phrases,...) to non-linguistic elements, such as the visual world (objects, people, ...)



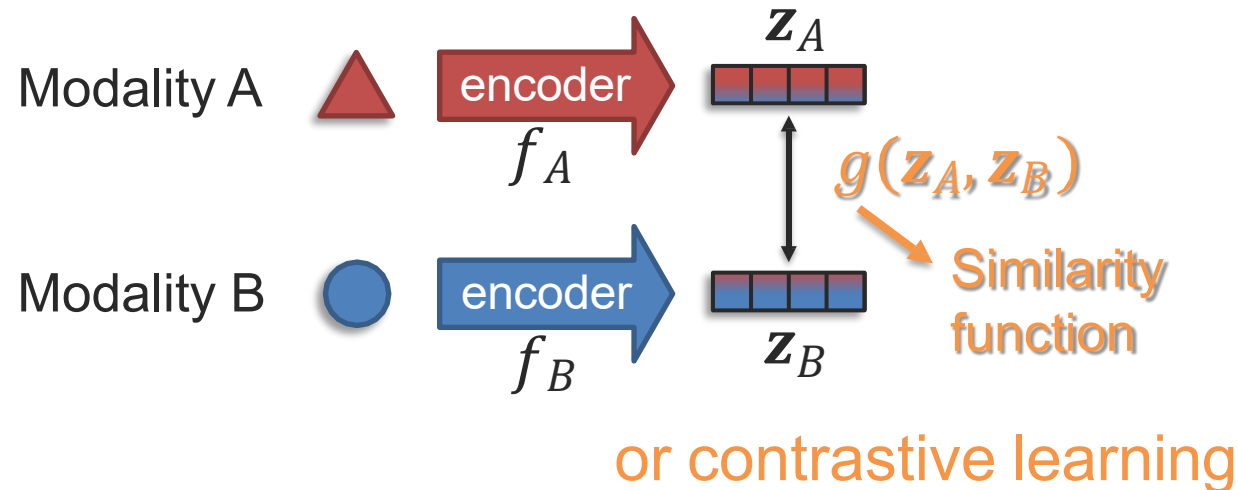
A **woman** reading **newspaper**

Local Alignment – Coordinated Representations

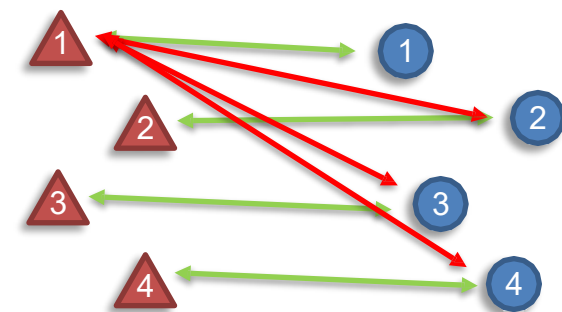


A **woman** reading **newspaper**

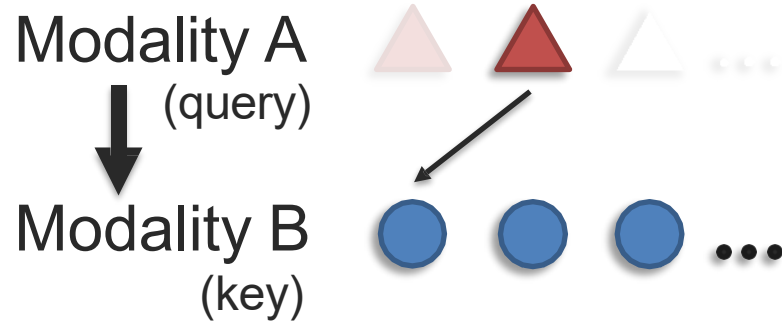
Learning coordinated representations:



Supervision: Paired data



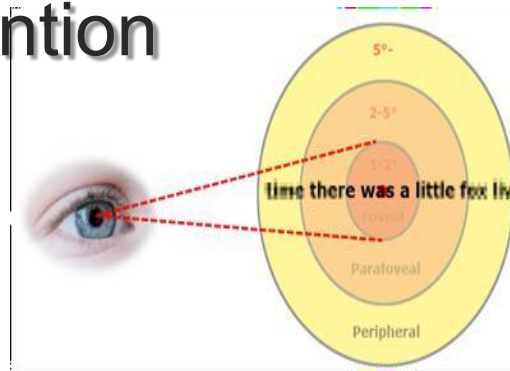
Directed Alignment



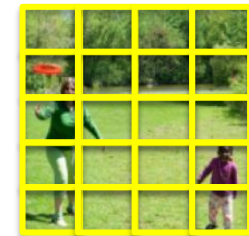
A woman is throwing a frisbee

Which
object?

Attention



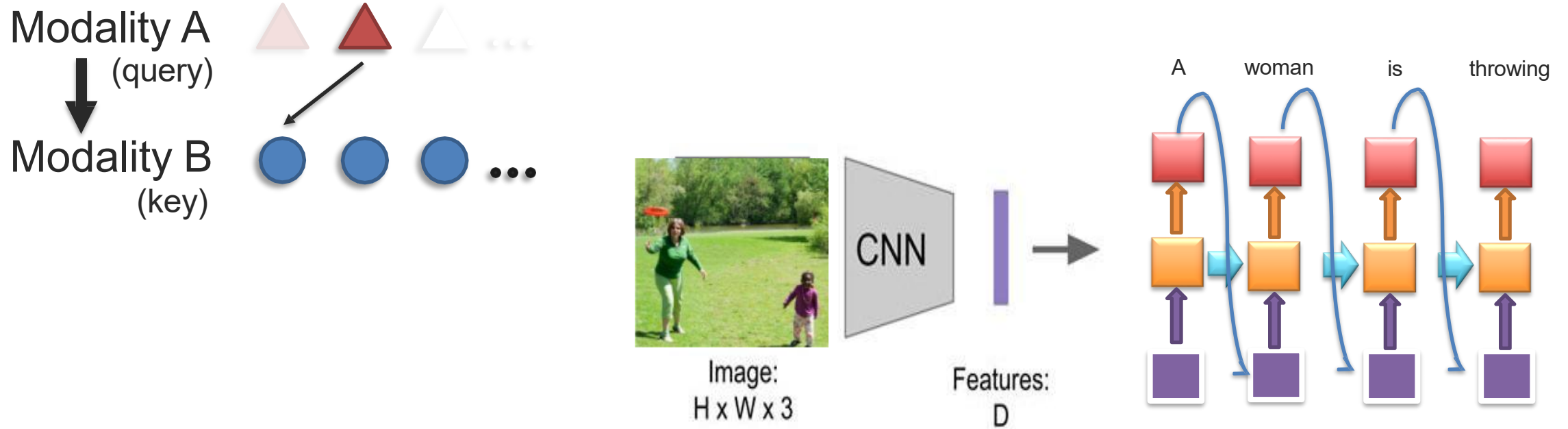
1 Soft attention



2 Hard attention

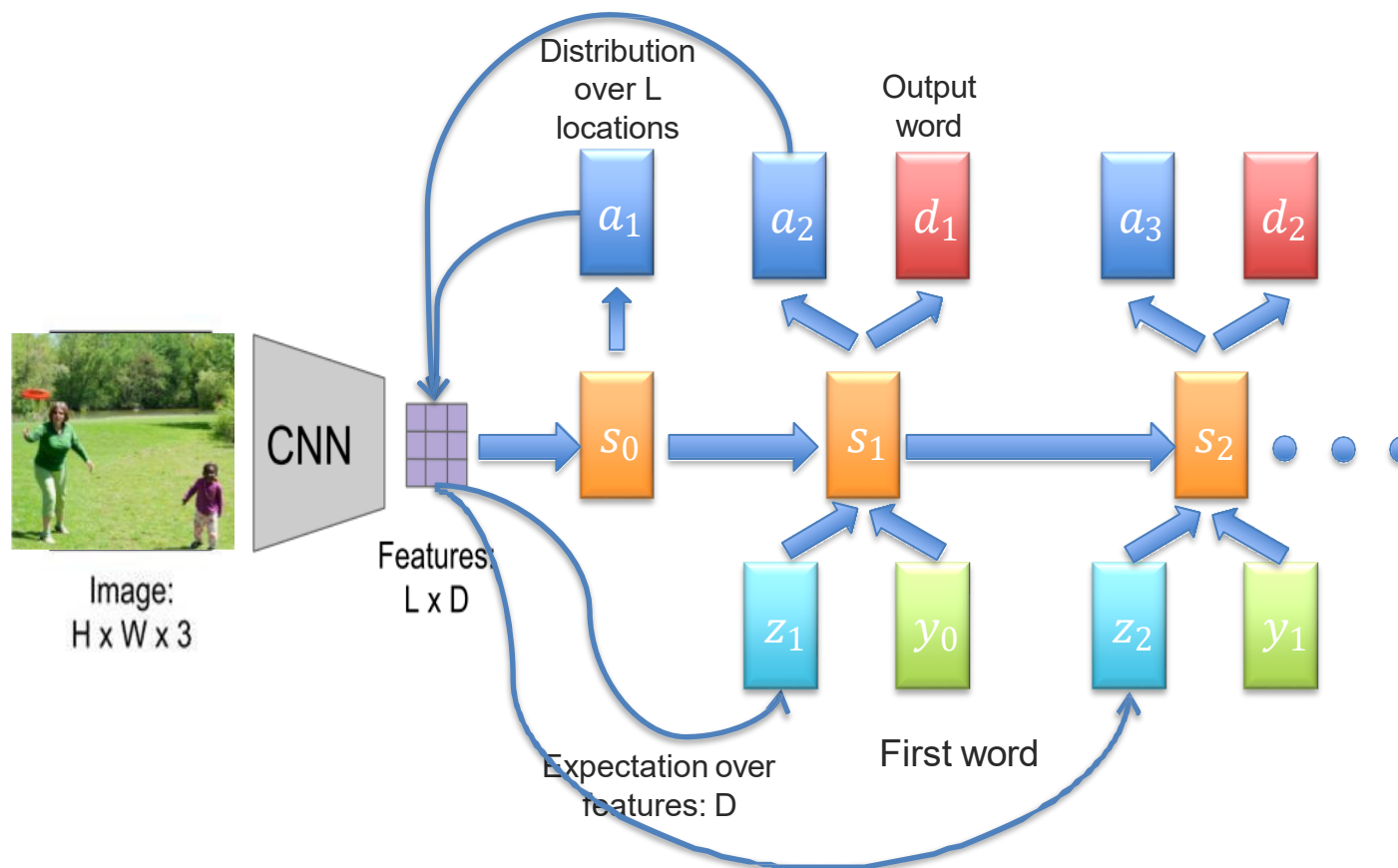


Directed Alignment – Image Captioning



Should we always use the final layer of the CNN for all generated words?

Directed Alignment – Image Captioning



Attention Gates

Before:

$$p(y_i|y_1, \dots, y_{i-1}, \mathbf{x}) = g(y_{i-1}, \mathbf{s}_i, \mathbf{z}),$$

where $\mathbf{z} = \mathbf{h}_T$, last encoder state and $\mathbf{s}_i$ is the current state of the decoder

Now:

$$p(y_i|y_1, \dots, y_{i-1}, \mathbf{x}) = g(y_{i-1}, \mathbf{s}_i, \mathbf{z}_i)$$

Have an attention “gate”

- A different context $\mathbf{z}_i$ used at each time step!
- $\mathbf{z}_i = \sum_{j=1}^{T_x} \alpha_{ij} \mathbf{h}_j$

α_{ij} is the (scalar) attention for word j at generation step i

Attention Gates

So how do we determine α_{ij} ?

$$\alpha_{i,j} = \frac{\exp(e_{ij})}{\sum_{k=1}^{T_x} \exp(e_{ik})} \quad \Rightarrow \text{softmax, making sure they sum to 1}$$

where:

$$e_{ij} = \mathbf{v}^T \sigma(W \mathbf{s}_{i-1} + U \mathbf{h}_j)$$

a feedforward network that can tell us how important the current encoding is

$\mathbf{v}$, W , U – learnable weights

$$\mathbf{z}_i = \sum_{j=1}^{T_x} \alpha_{ij} \mathbf{h}_j$$

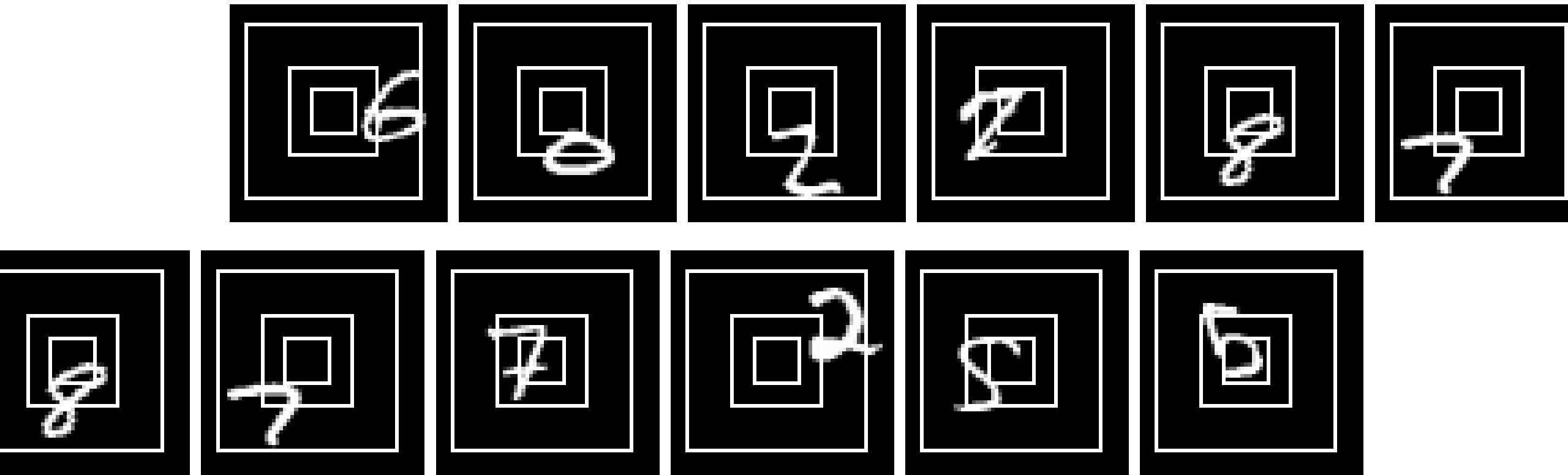
← expectation of the context (a fancy way to say it's a weighted average)

Example – Image Captioning

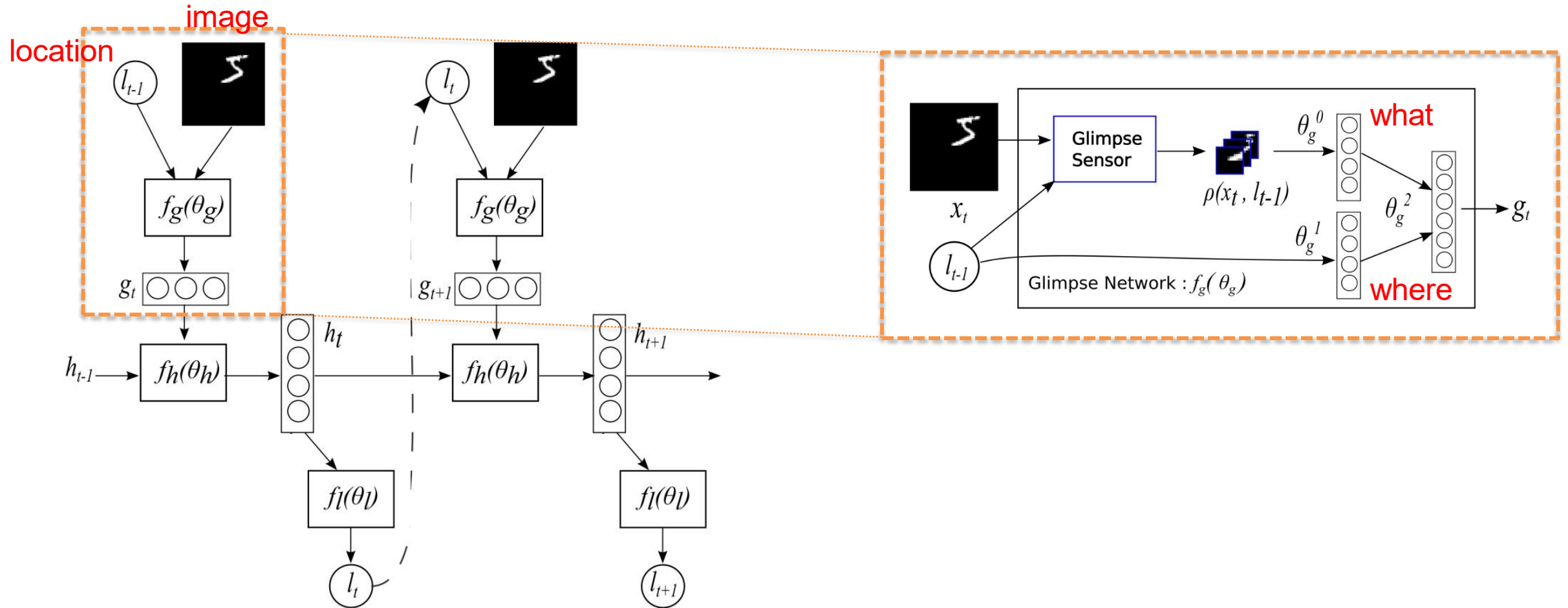


[Show, Attend and Tell: Neural Image Caption Generation with Visual Attention, Xu et al., 2015]

Hard attention - Example



Hard Attention – Recurrent Model of Visual Attention



Global Alignment

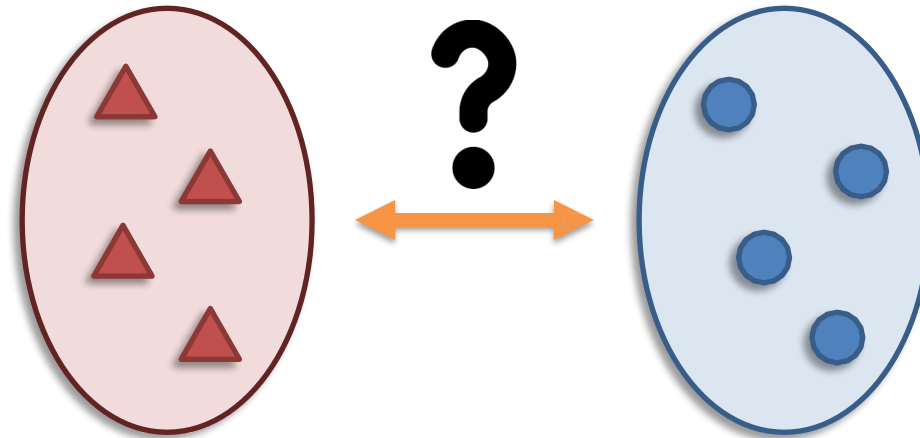
Visual



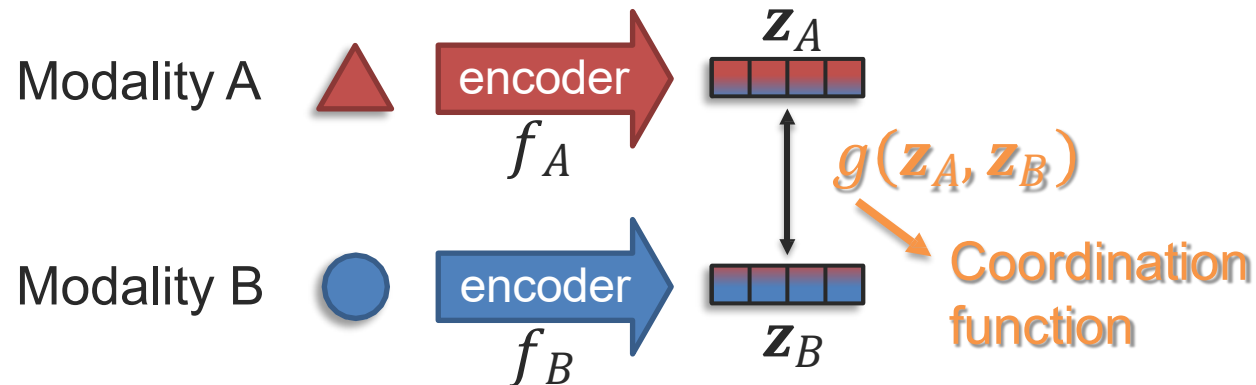
Language



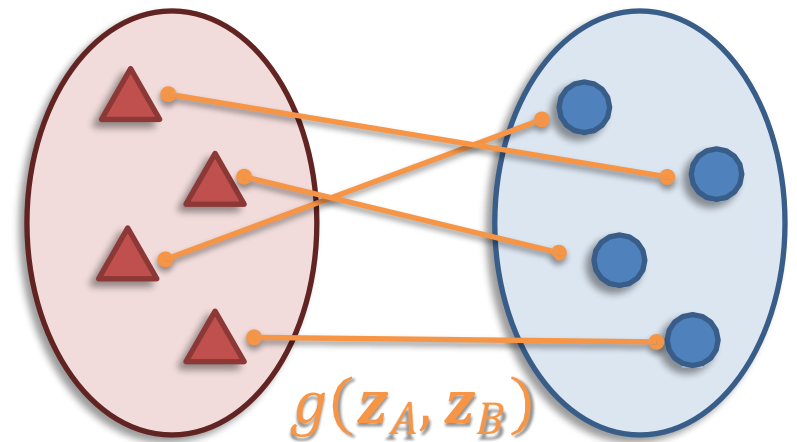
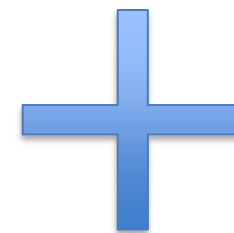
Latent pairing information



Jointly optimize representation + global alignment:

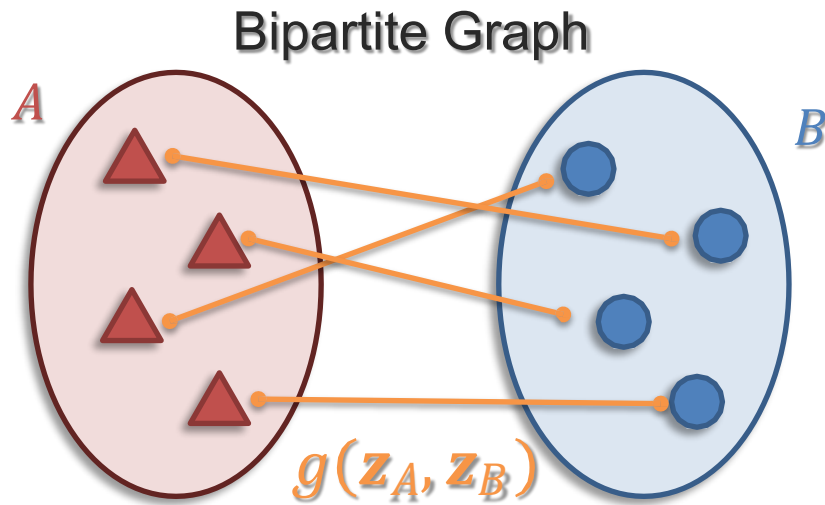


(representation)



(global alignment)

Assignment Problem



Initial assumptions:

- Same number of elements in A and B modalities
- 1-to-1 “hard” alignment between elements
- All elements assigned (aka “perfect matching”)

➡ How to solve?

Naive solution: check all assignments

Better solution: Linear Programming

Assignment: ~~$f: A \rightarrow B$~~
(vector of indices)

Similarity weights: ~~$w_{(i, f(i))} = g(\mathbf{z}_A^i, \mathbf{z}_B^{f(i)})$~~

Maximize: ~~$\max_{f \in \text{Perm}(N)} \sum_{i=1}^N w_{i, f(i)}$~~

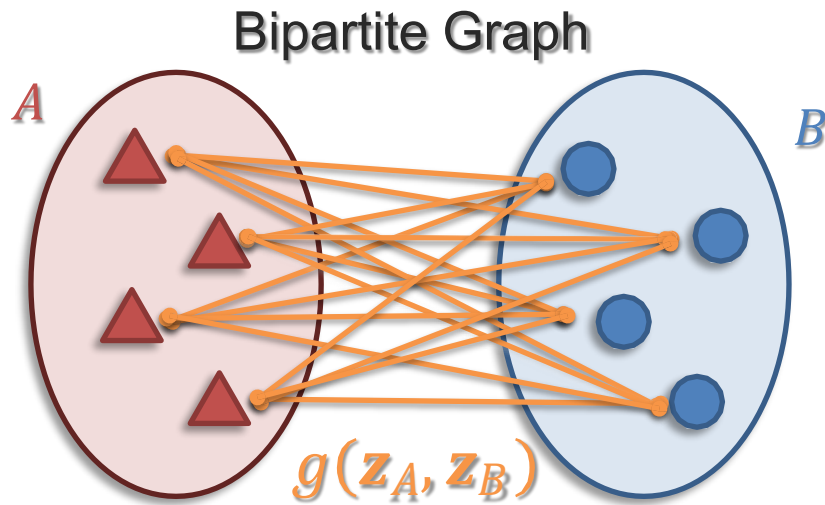
$x_{ij} = 1$ when matching connection, otherwise 0

$w_{(i,j)} = g(\mathbf{z}_A^i, \mathbf{z}_B^j)$

$\max_{\{x_{ij}\}} \sum_{(i,j) \in A \times B} w_{i,j} x_{ij}$

➡ Can be solved with simplex algorithm

Optimal transport



New assumptions:

- Different number of elements in A and B modalities
- Many-to-many “soft” alignment between elements

It can be seen as “transporting” elements from modality A to modality B (and vice-versa)

Assignments: $x_{(i,j)}$: soft alignment between $\mathbf{z}_A^i$ and $\mathbf{z}_B^j$

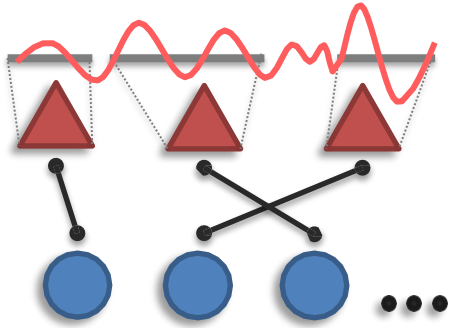
Similarity weights: $w_{(i,j)} = g(\mathbf{z}_A^i, \mathbf{z}_B^j)$

Maximize:
$$\max_{\{x_{ij}\}} \sum_{(i,j) \in A \times B} w_{i,j} x_{ij}$$

➡ Wassertein distance
gives optimal transport

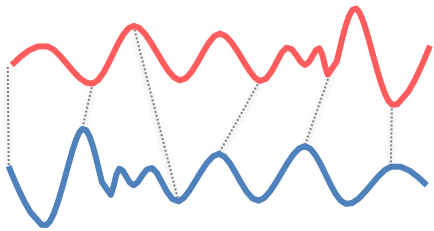
Continuous Alignment

Challenge 2b: Continuous Alignment

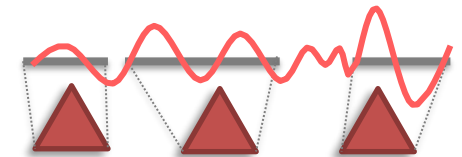


Definition: Model alignment between modalities with continuous signals and no explicit elements

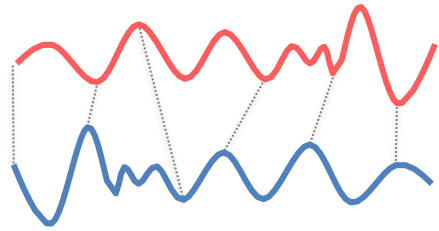
Continuous
warping



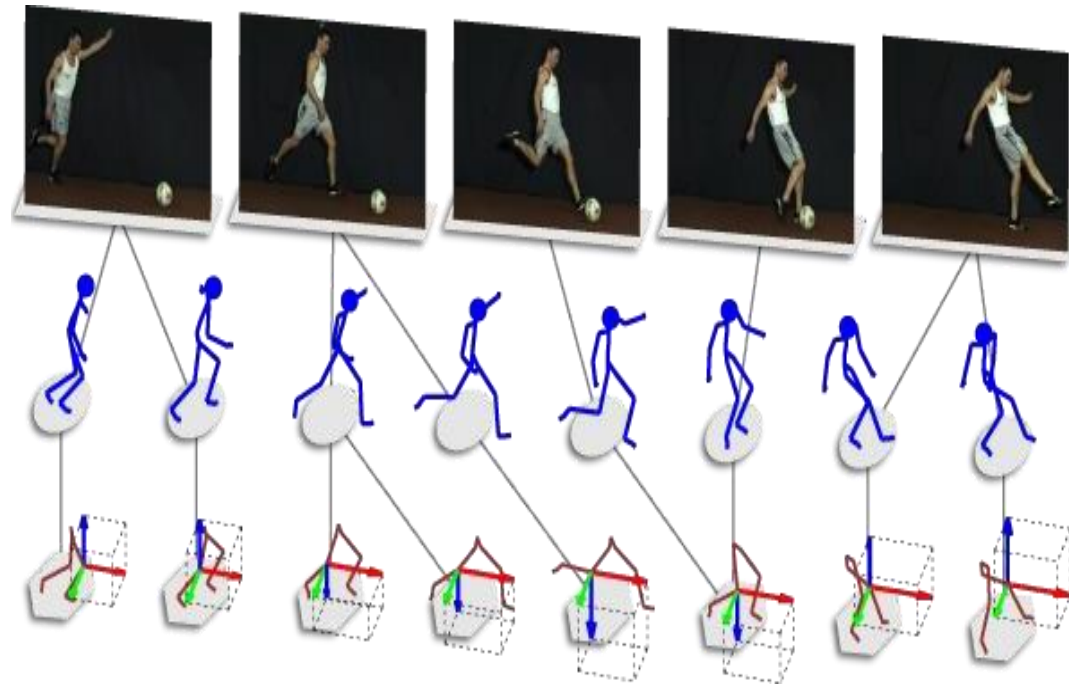
Discretization
(segmentation)



Continuous Warping – Example



➡ Aligning video sequences



Dynamic Time Warping (DTW)

We have two unaligned temporal unimodal signals

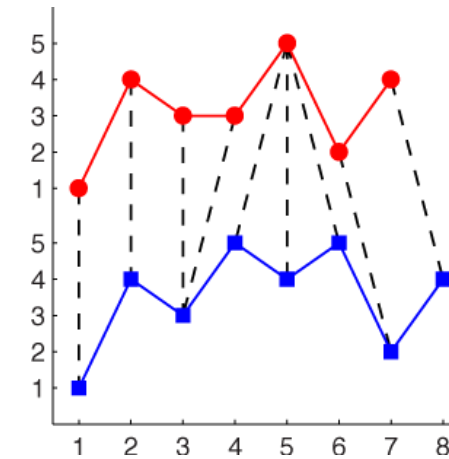
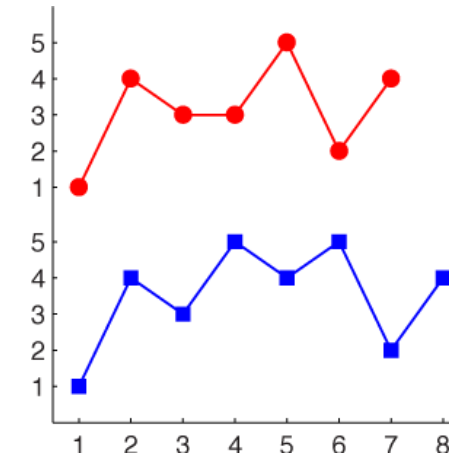
- $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_{n_x}] \in \mathbb{R}^{d \times n_x}$
- $\mathbf{Y} = [\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_{n_y}] \in \mathbb{R}^{d \times n_y}$

Find set of indices to minimize the alignment difference:

$$L(\mathbf{p}^x, \mathbf{p}^y) = \sum_{t=1}^l \|\mathbf{x}_{p_t^x} - \mathbf{y}_{p_t^y}\|_2^2$$

where $\mathbf{p}^x$ and $\mathbf{p}^y$ are index vectors of same length

Dynamic Time Warping is designed to find these index vectors!



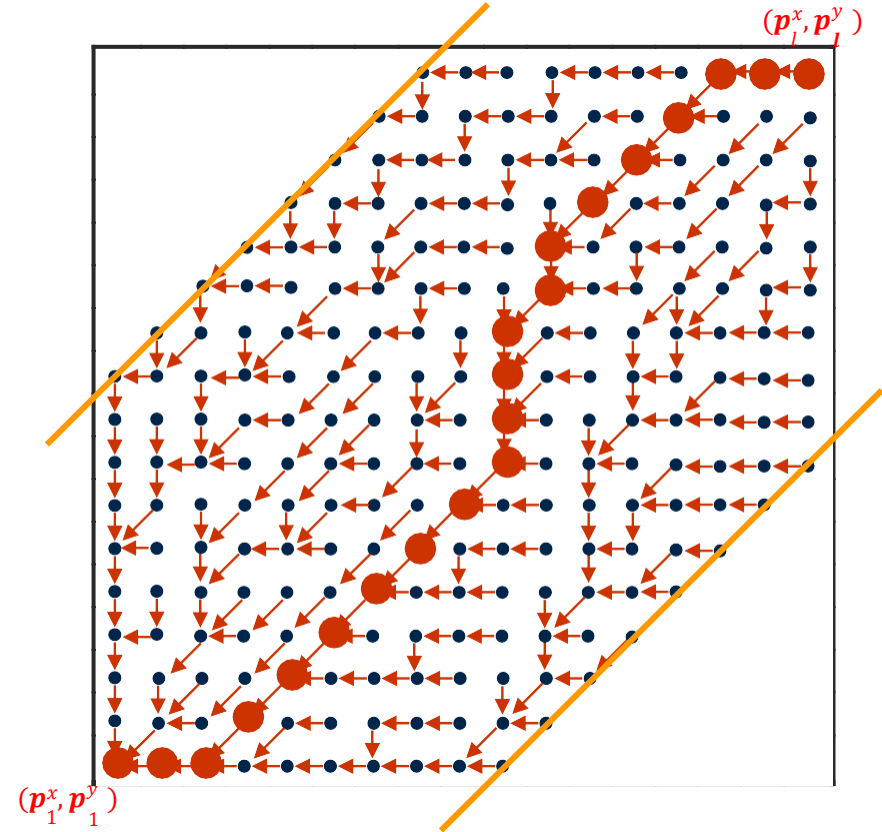
Dynamic Time Warping (DTW)

Lowest cost path in a cost matrix

- Restrictions?

- Monotonicity – no going back in time
- Continuity - no gaps
- Boundary conditions - start and end at the same points
- Warping window - don't get too far from diagonal
- Slope constraint – do not insert or skip too much

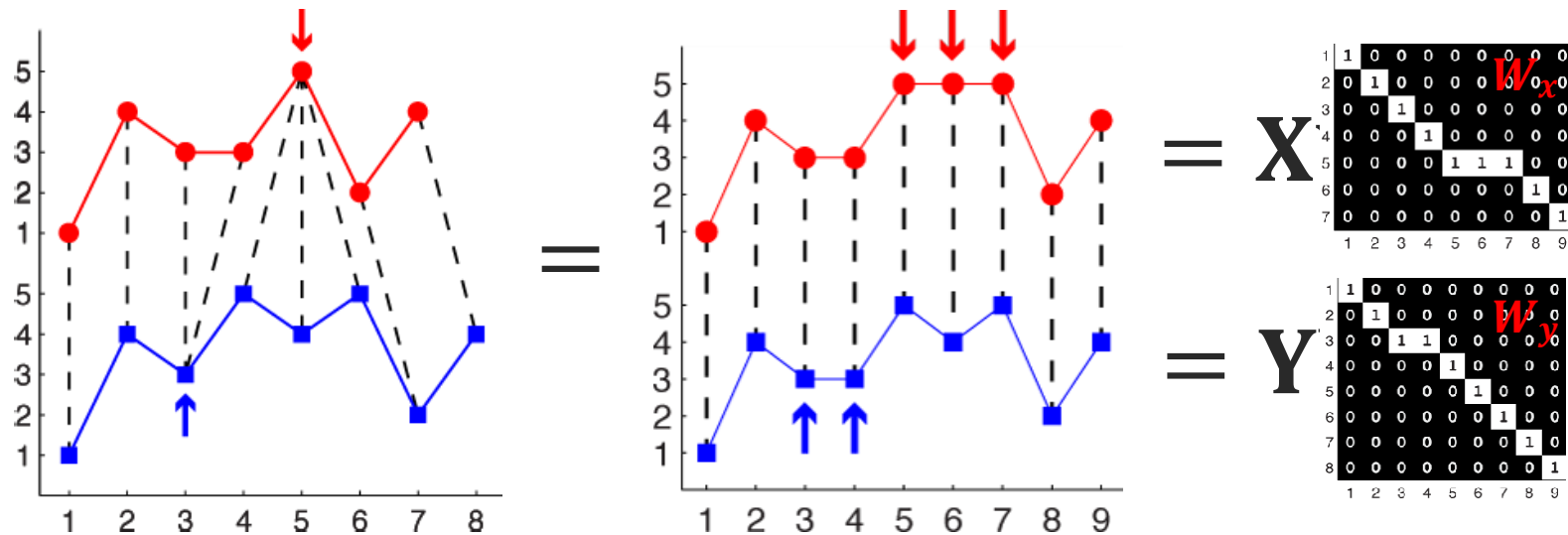
Solved using dynamic programming while respecting the restrictions



DTW alternative formulation

$$L(\mathbf{p}^x, \mathbf{p}^y) = \sum_{t=1}^l \left\| \mathbf{x}_{\mathbf{p}_t^x} - \mathbf{y}_{\mathbf{p}_t^y} \right\|_2^2$$

Replication doesn't change the objective!



Alternative objective:

$$L(W_x, W_y) = \left\| XW_x - YW_y \right\|_F^2$$

Frobenius norm $\|A\|_F^2 = \sum_i \sum_j |a_{i,j}|^2$

X, Y – original signals (same #rows, possibly different #columns)

W_x, W_y – alignment matrices

A differentiable version of DTW also exists...

<https://arxiv.org/pdf/1703.01541.pdf>

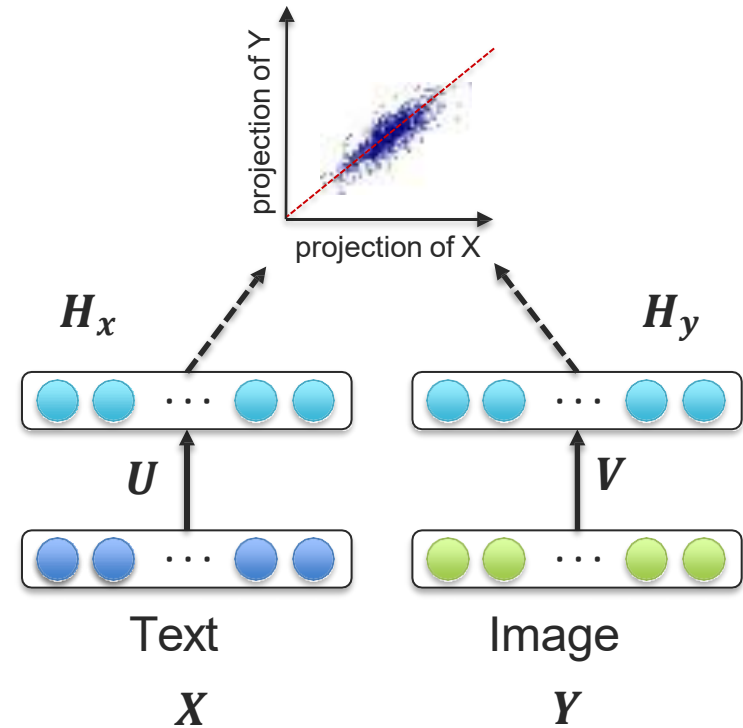
Canonical Correlation Analysis – Reminder

CCA loss can also be re-written as:

$$L(U, V) = \|U^T \mathbf{X} - V^T \mathbf{Y}\|_F^2$$

subject to:

$$U^T \Sigma_{YY} U = V^T \Sigma_{YY} V = I, u_{(j)}^T \Sigma_{XY} v_{(i)} = 0$$



Canonical Time Warping

Dynamic Time Warping + Canonical Correlation Analysis = Canonical Time Warping

$$L(\mathbf{U}, \mathbf{V}, \mathbf{W}_x, \mathbf{W}_y) = \|\mathbf{U}^T \mathbf{X} \mathbf{W}_x - \mathbf{V}^T \mathbf{Y} \mathbf{W}_y\|_F^2$$

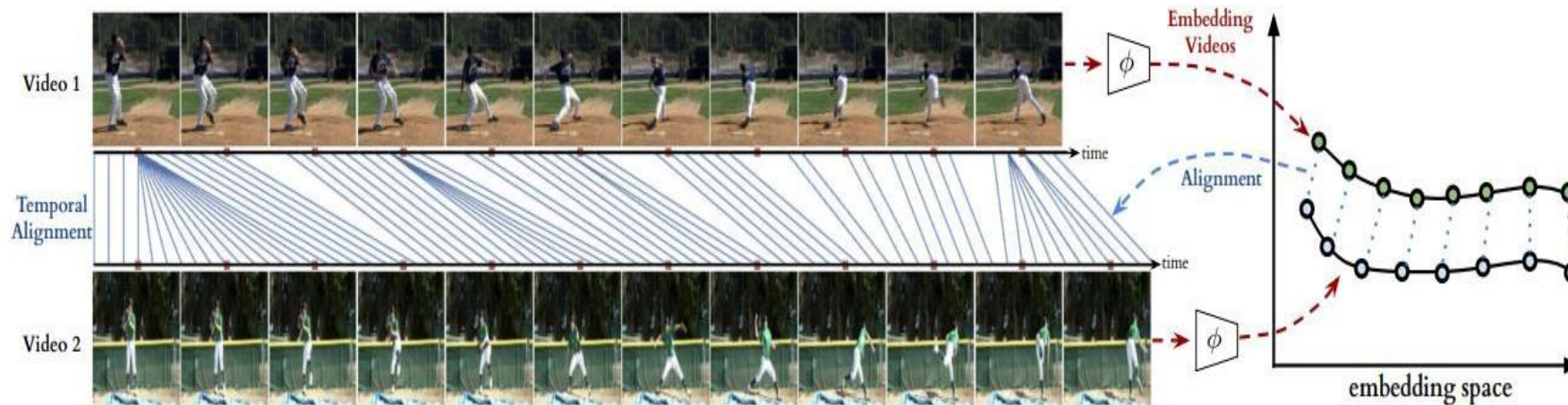
Allows to align multi-modal or multi-view (same modality but from a different point of view)

- $\mathbf{W}_x, \mathbf{W}_y$ – temporal alignment
- $\mathbf{U}, \mathbf{V}$ – cross-modal (spatial) alignment

[Canonical Time Warping for Alignment of Human Behavior, Zhou and De la Tore, 2009]

Temporal Alignment and Neural Representation Learning

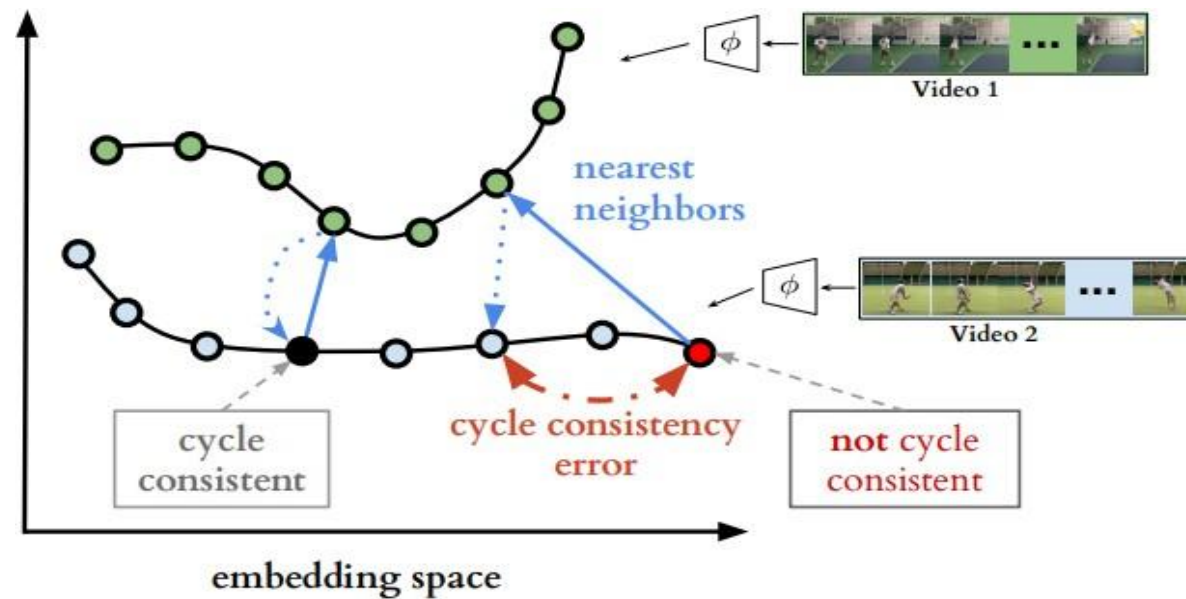
Premise: we have paired video sequences that can be temporally aligned



How can we define a loss function to enforce the alignment between sequences while at the same time learning good representations?

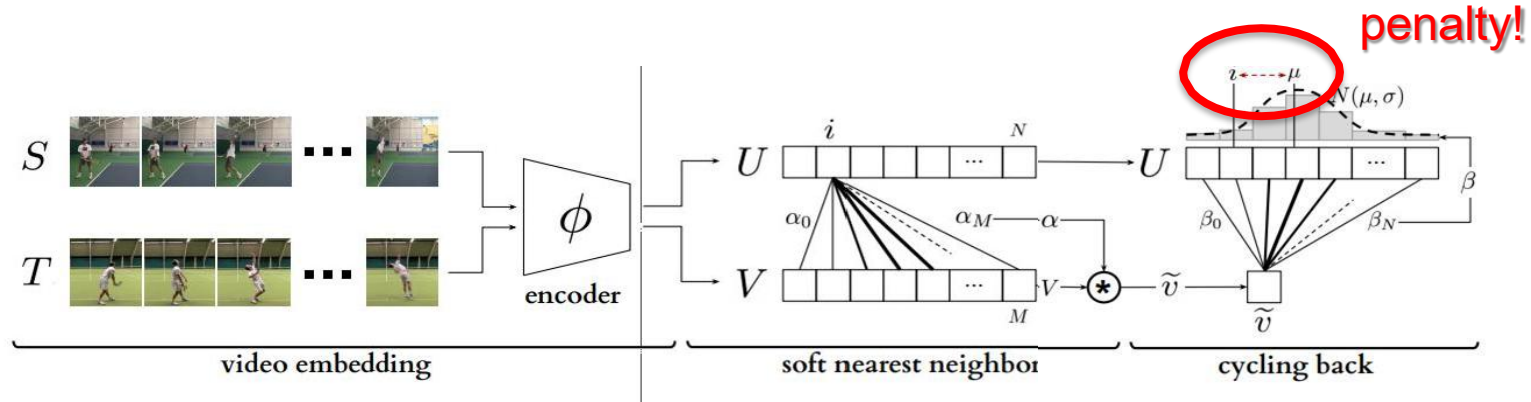
Temporal Cycle-Consistency Learning

Solution: Representation learning by enforcing **Cycle consistency**



Main idea: My closest neighbor also views me as their closest neighbor

Temporal Cycle-Consistency Learning



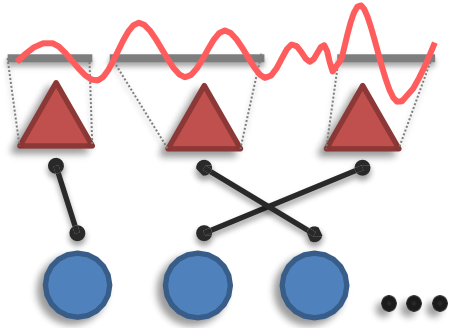
Compute “soft” / “weighted” nearest neighbour:

distances: $\alpha_j = \frac{e^{-||u_i - v_j||^2}}{\sum_k^M e^{-||u_i - v_k||^2}}$ Soft nearest neighbor: $\tilde{v} = \sum_j^M \alpha_j v_j,$

Find the nearest neighbor the other way and then penalize the distance:

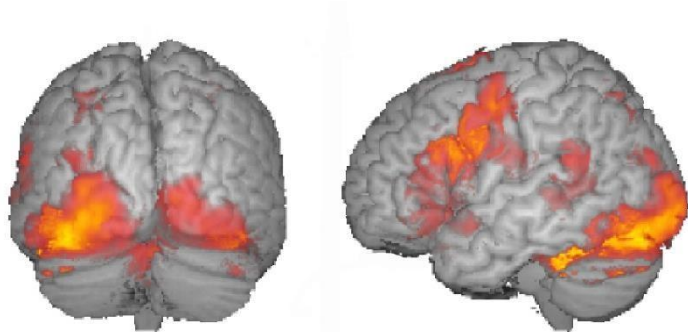
$$\beta_k = \frac{e^{-||\tilde{v} - u_k||^2}}{\sum_j^N e^{-||\tilde{v} - u_j||^2}} \quad L_{cbr} = \frac{|i - \mu|^2}{\sigma^2} + \lambda \log(\sigma)$$

Discretization (aka Segmentation)

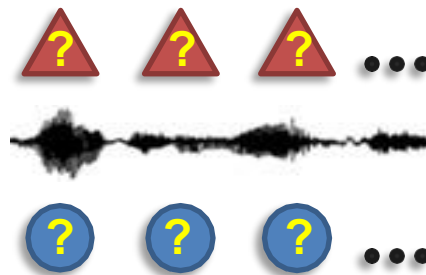


Common assumptions: ① Segmented elements

Examples:



Medical imaging



Signals

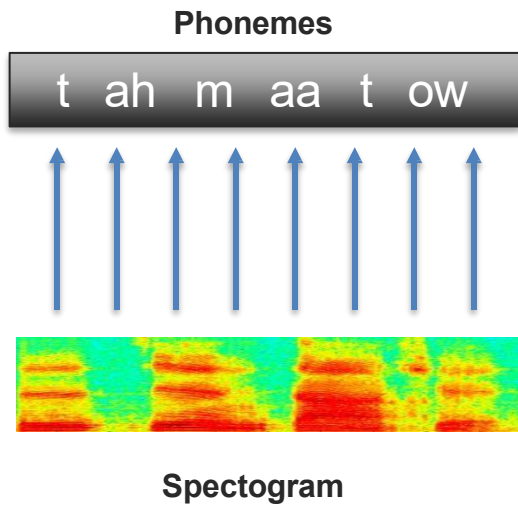


Images

objects

Discretization – Example

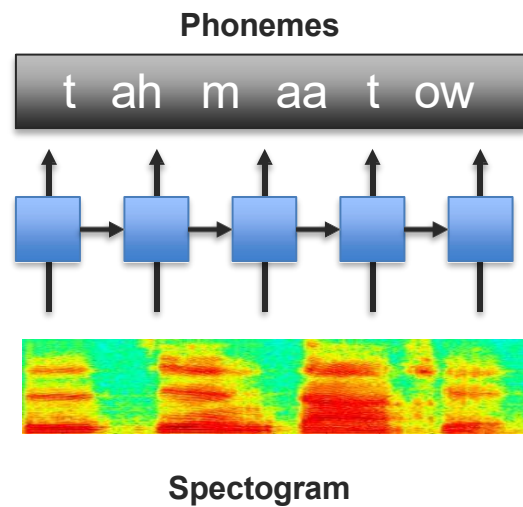
Sequence Labeling and Alignment



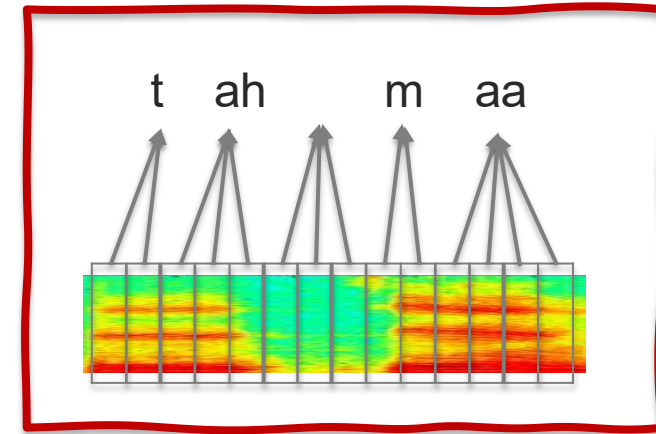
How can we predict the sequence
of phoneme labels?

Discretization – Example

Sequence Labeling and Alignment



Challenge: many-to-1 alignment

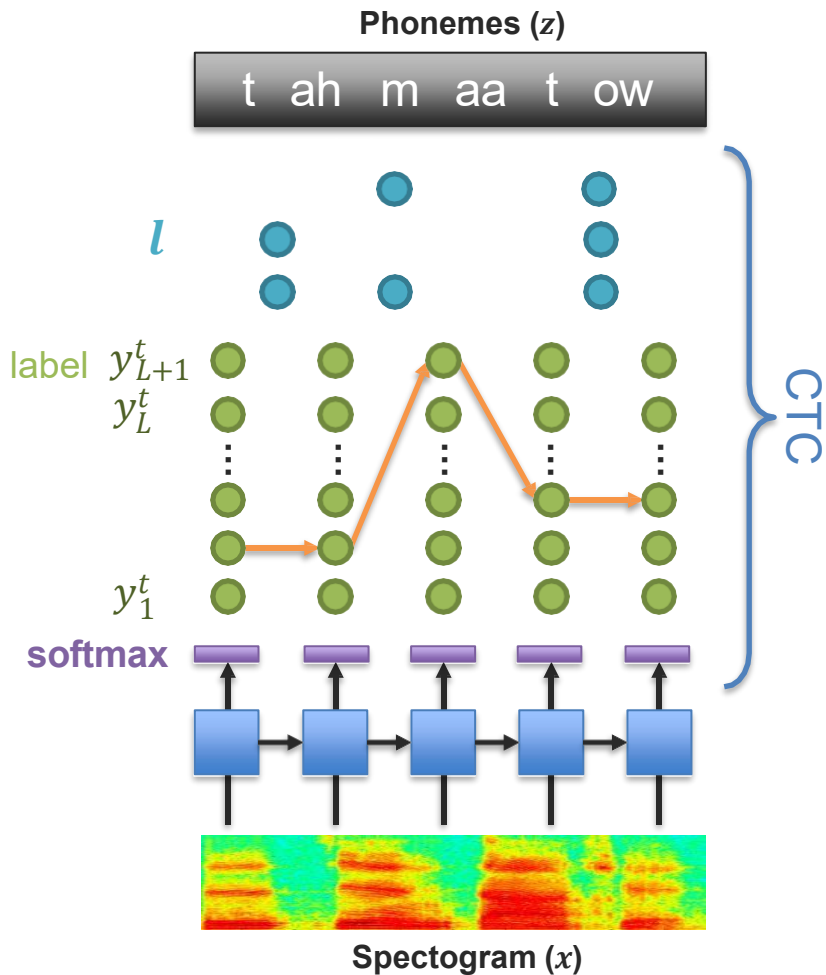


How can we predict the sequence
of phoneme labels?

Discretization – A Classification Approach

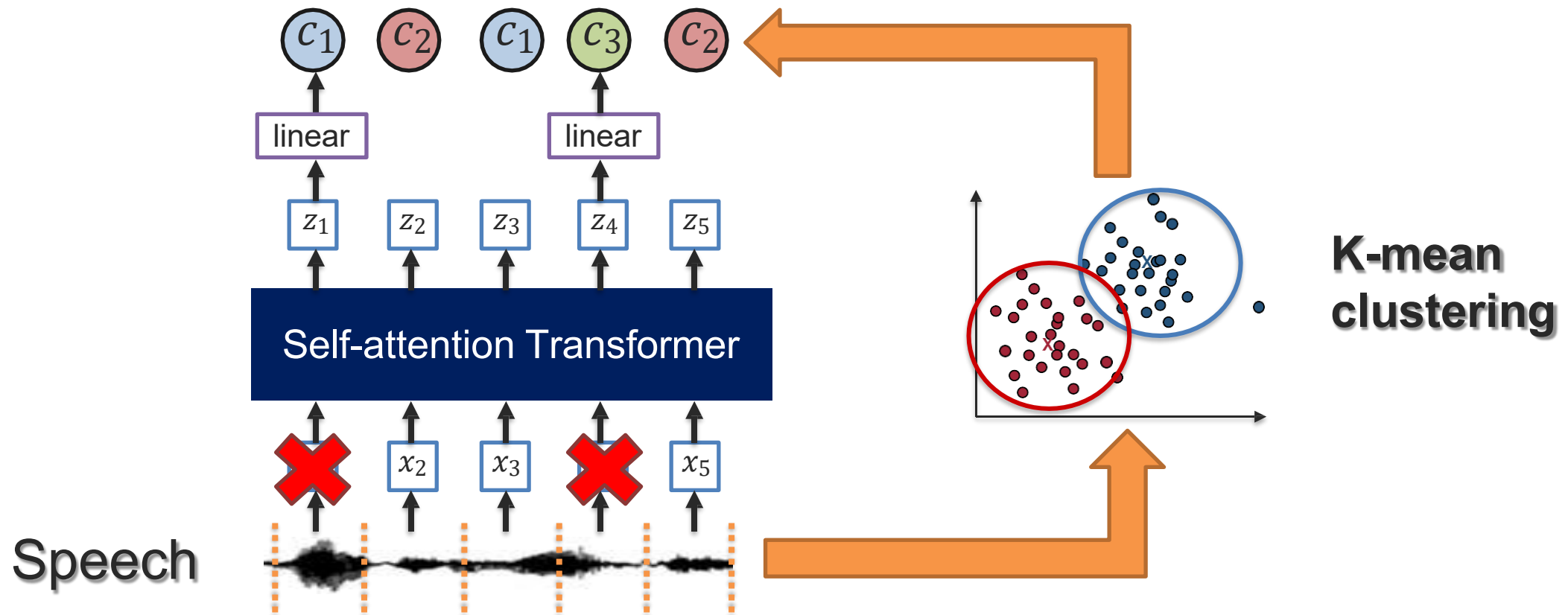
Connectionist Temporal Classification

- ④ Most probable sequence labels
- ③ Predicted labels l
- ② Path π over the activations:
- ① Output activations (distribution):



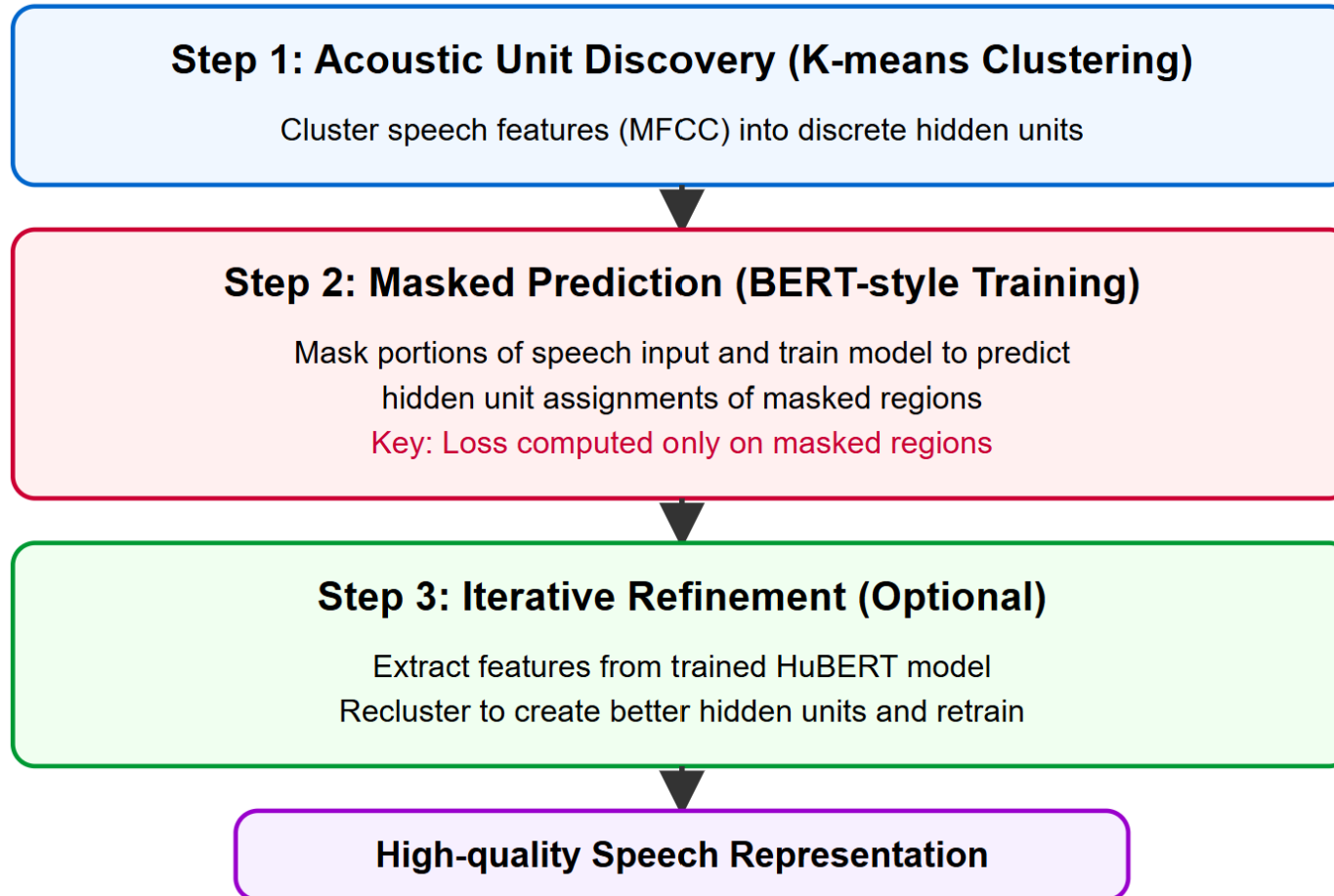
Discretization and Representation – Cluster-based Approaches

HUBERT: Hidden-Unit BERT



Discretization and Representation – Cluster-based Approaches

HuBERT: Hidden-Unit BERT Approach

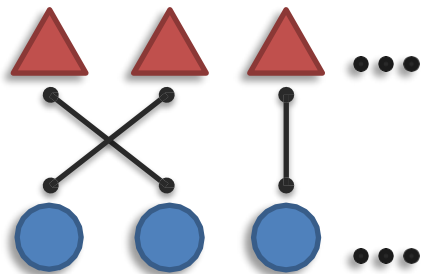


Challenge 2: Alignment

Definition: Identifying and modeling cross-modal connections between all elements of multiple modalities, building from the data structure

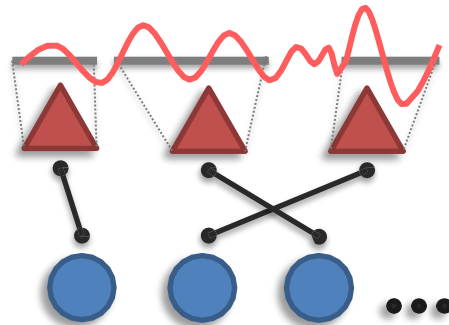
Sub-challenges:

Discrete Alignment



Discrete elements
and connections

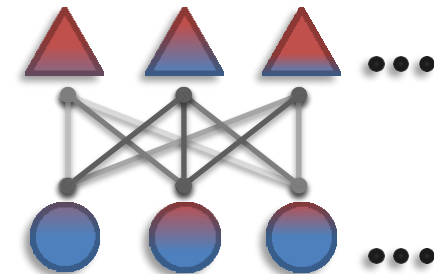
Continuous Alignment



Segmentation and
continuous warping

Next!

Contextualized Representation



Alignment
+ representation

References

- Multi Modal Machine Learning, 11-777 • Fall 2023 • Carnegie Mellon University

Supplement: deeper detail on multimodal alignment

Added examples, trade-offs, and method-selection
guidance

Unique, shared, and synergistic information

Unique information



$$I(A;Y|B)$$

Only one modality contributes the critical clue.

Example: radar uniquely estimates distance in fog.

Shared / redundant

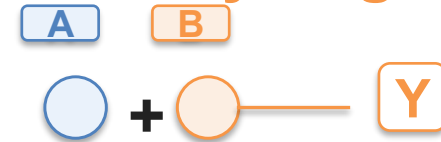


Redundancy R

Both modalities carry similar task signal.

Example: lip motion and audio both cue the phoneme.

Synergistic



Synergy S

The task becomes clear only when signals are combined.

Example: words + prosody reveal sarcasm.

Design implication: redundancy improves robustness, while synergy is where cross-modal interaction and fusion matter most.

Source: Williams & Beer (2010) on partial information decomposition; examples synthesized for teaching.

CLIP-style contrastive alignment inside one batch

Similarity matrix $S(i,j) = \cos(z_i^I, z_j^T)/\tau$

		text embeddings				
		txt1	txt2	txt3	txt4	txt5
enc	text encoder					
img1		+	-	-	-	-
img2		-	+	-	-	-
img3		-	-	+	-	-
img4		-	-	-	+	-
img5		-	-	-	-	+
enc	image encoder					

1. Training signal

Diagonal entries are the positives; off-diagonals in the batch are negatives.
The loss is usually symmetric: image→text and text→image.

2. Why it works well

Large batches give many hard negatives “for free”.
Temperature τ forces the true pair to stand out.

3. What it does not directly give you

CLIP aligns whole images with whole captions.
Fine-grained region or token grounding needs extra machinery.

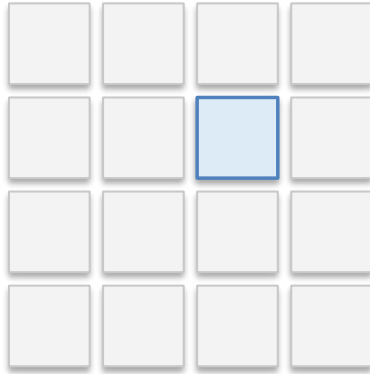
Key intuition: contrastive learning teaches a ranking, not an explicit point-to-point correspondence.

Directed alignment with attention: hard vs soft

Hard attention

query

"frisbee"



one sampled
location

selection

One discrete glimpse / region

training

Training usually needs sampling

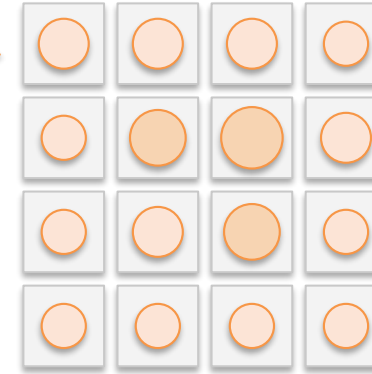
use case

Useful when compute is limited

Soft attention

query

"frisbee"



weighted
average
of all locations

selection

Distribution over all locations

training

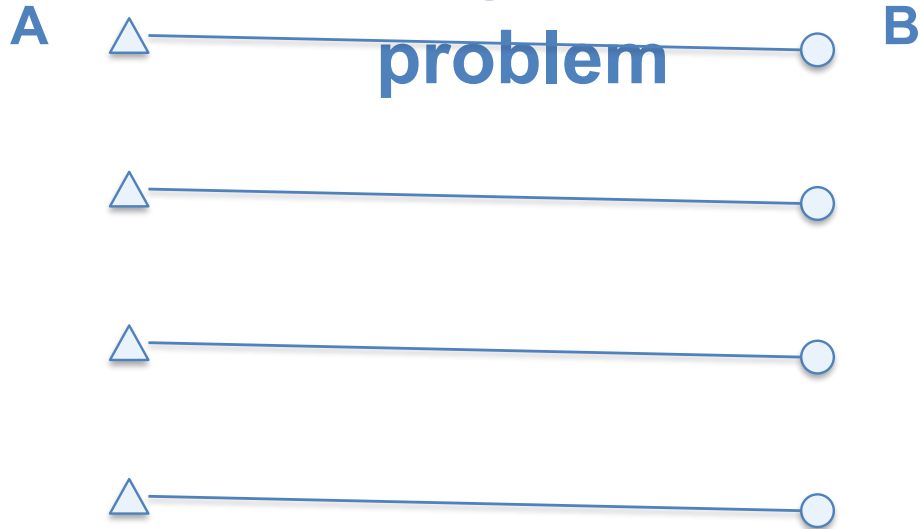
Fully differentiable end-to-end

use case

Default in seq2seq models and transformers

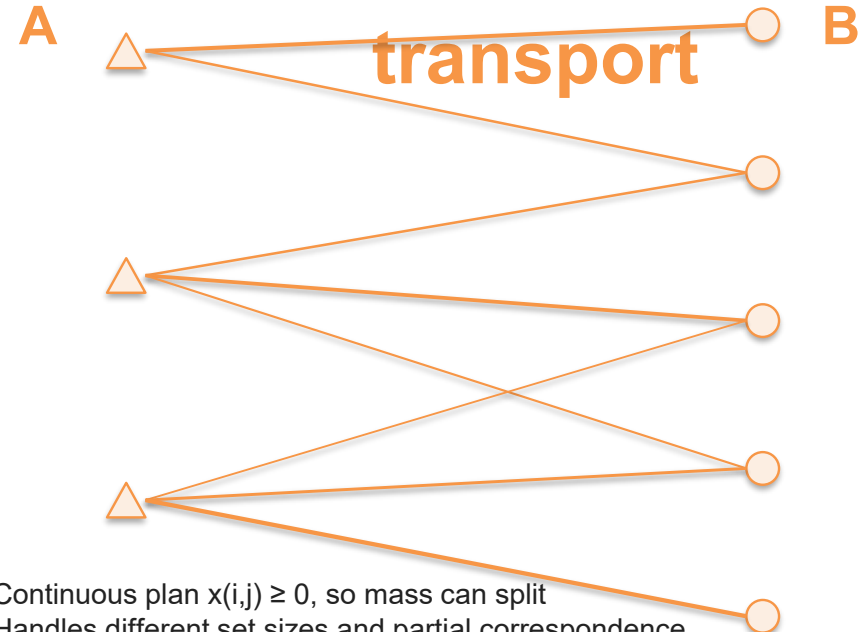
From assignment to optimal transport

Assignment problem



Binary decision $x(i,j) \in \{0,1\}$
Best when every element needs exactly one partner
Typical for equal-size sets and hard 1-to-1 matching

Optimal transport



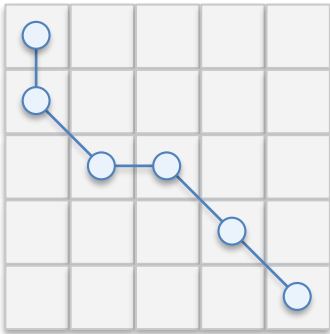
Continuous plan $x(i,j) \geq 0$, so mass can split
Handles different set sizes and partial correspondence
Useful when alignment should be soft or many-to-many

Rule of thumb: need one exact partner for each element → assignment. Need partial, weighted, or unequal matching → OT.

Sources: Kuhn (1955) for the assignment / Hungarian perspective; Peyré & Cuturi (2019) for optimal transport.

The DTW family: DTW, Soft-DTW, and CTW

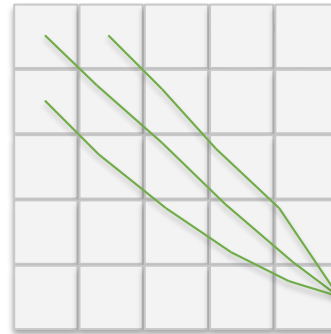
DTW



One best path through the cost matrix

Aligns variable-speed sequences
Returns one optimal path
Best when timing is the main issue

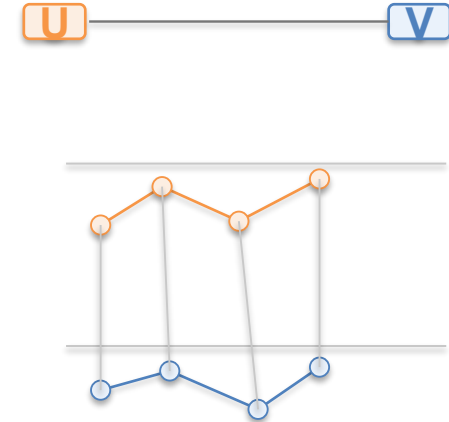
Soft-



Smooth min → differentiable loss

Keeps the DTW intuition
Useful inside trainable models
Often used as a differentiable loss

CTW



Project features before warping

Helps when feature spaces differ
Learns U,V with the alignment
Bridges time warping and multimodal alignment

Use DTW for path alignment, Soft-DTW for differentiable training, and CTW when alignment must also reconcile feature spaces.

Sources: Sakoe & Chiba (1978) for DTW, Cuturi & Blondel (2017) for Soft-DTW, Zhou & de la Torre (2009) for Canonical Time Warping

CTC collapse rule: from frame scores to labels

One valid frame-level path π

1. Per-frame path emitted by the network

∅ t t ah ∅ m aa aa t ow ∅

merge
repeats

2. Collapse consecutive duplicates

∅ t ah ∅ m aa t ow ∅

remove
blanks

3. Final label sequence

t ah m aa t ow

Why CTC helps

**No pre-segmented
boundaries needed**

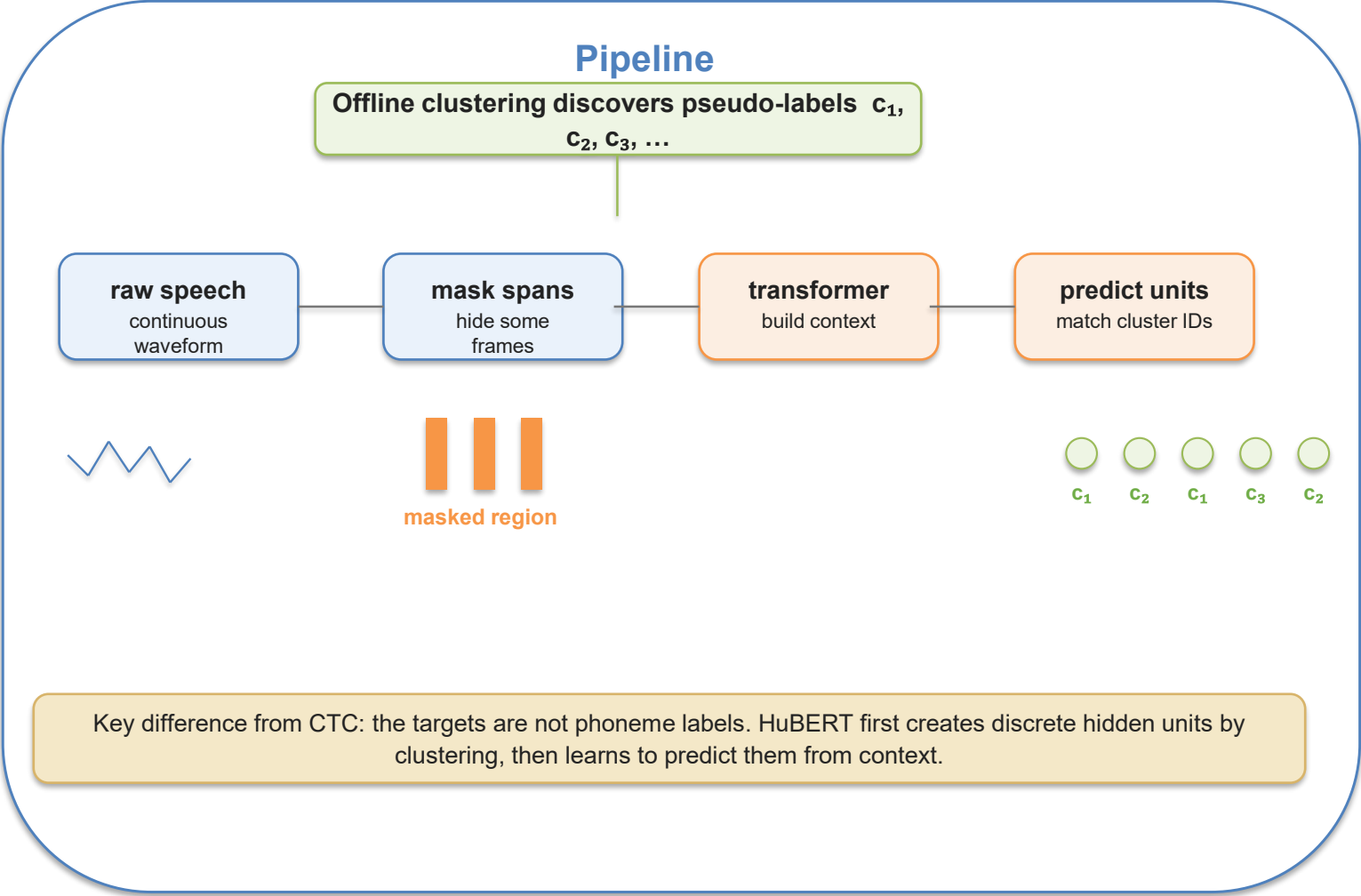
Per-time-step distributions
instead of explicit cuts

Sums over all paths that
collapse to the same label
sequence

The blank token absorbs
duration differences

That is why CTC solves the
“many frames → one label”
problem elegantly.

Cluster-based discretization: what HuBERT adds



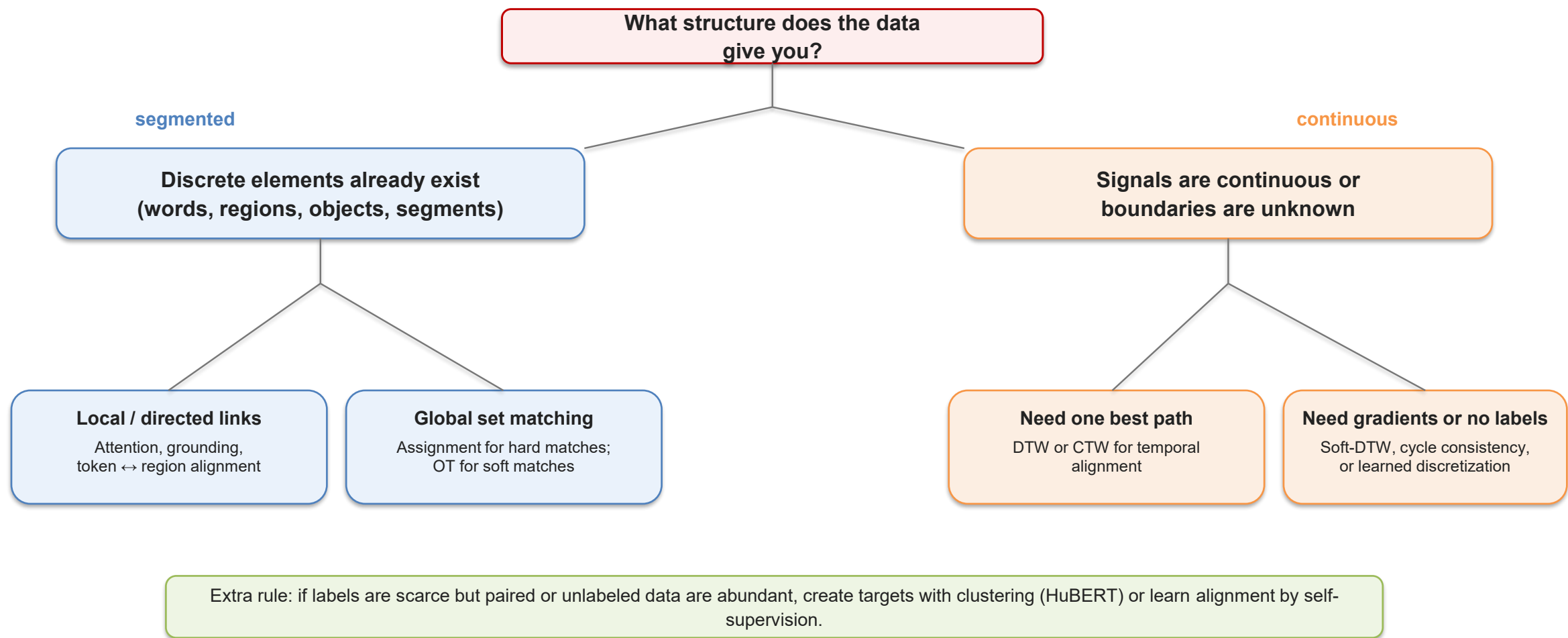
Why it matters for alignment

Speech becomes a more stable sequence of units.
The units come from unlabeled audio via clustering.
Useful for representation learning, not just decoding.

Supervised path: CTC
Self-supervised path: HuBERT

Source: Hsu et al. (2021), HuBERT: Self-Supervised Speech Representation Learning by Masked Prediction of Hidden Units.

Choosing the right alignment strategy



Supplementary references

Foundations

Williams, P. L., & Beer, R. D. (2010). Nonnegative decomposition of multivariate information.
van den Oord, A., Li, Y., & Vinyals, O. (2018). Representation learning with contrastive predictive coding.
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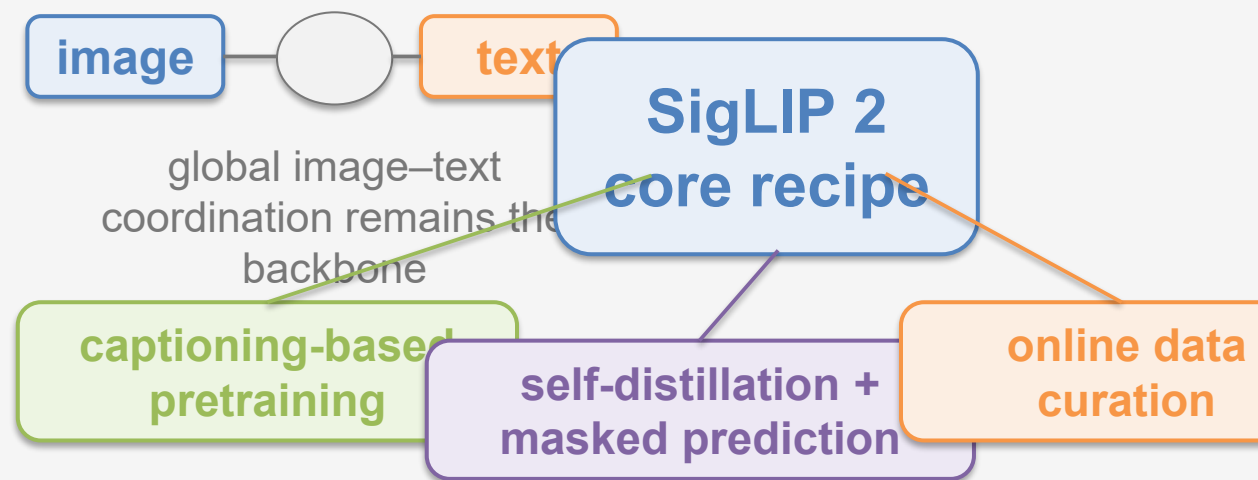
Alignment & discretization

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Peyré, G., & Cuturi, M. (2019). Computational optimal transport.
Sakoe, H., & Chiba, S. (1978). Dynamic programming algorithm optimization for spoken word recognition.
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Cuturi, M., & Blondel, M. (2017). Soft-DTW: A differentiable loss function for time-series.
Graves, A. et al. (2006). Connectionist temporal classification.
Hsu, W.-N. et al. (2021). HuBERT: Self-supervised speech representation learning by masked prediction of hidden units.

Recent developments (2024–2026)

How classic alignment ideas reappear inside
today's foundation models

Better image–text encoders: SigLIP 2



What changes vs. classic CLIP?

- Outperforms earlier SigLIP models on zero-shot classification, image–text retrieval, and transfer.
- Improves localization and dense prediction, not only global matching.
- Supports multiple resolutions and preserves native aspect ratio.

Lesson: auxiliary objectives turn image–text embeddings into stronger grounding features.

Source: Tschannen et al., SigLIP 2, 2025.

Inside MLLMs: token-level and hidden-state alignment

SEA: Supervised Embedding Alignment

visual
tokens

LLM / text
space

Contrastive token-level supervision during pretraining

- Image-level supervision often leaves a mismatch between visual tokens and the language model.
- SEA aligns visual tokens directly to the LLM embedding space for more precise visual-text integration.
- Smaller models benefit the most because they have less spare capacity to repair bad alignment later.

VIRAL: Visual Representation Alignment

vision
features

MLLM middle
visual states

Regularize hidden visual states instead of relying on text-only supervision

- Visual instruction tuning can discard fine-grained detail on the way to the LLM.
- VIRAL aligns hidden visual representations with strong pretrained vision foundation features.
- This improves counting, spatial reasoning, and visually grounded responses.

Trend: modern MLLMs align at token level and hidden-state level inside the model.

Sources: Yin et al., SEA, 2025; Yoon et al., VIRAL, 2025.

Temporal / omni-modal alignment: hard but improving

MMAAlign: in open-source OLLMs, audio ↔ vision alignment is still weaker than text ↔ vision.

Qwen2.5-VL

stronger video / document alignment

- Dynamic resolution for images of varying sizes.
- Precise object localization and robust document parsing.
- Long-video comprehension with second-level event localization.

Qwen2.5-Omni

synchronize audio + video in one model

- End-to-end text, image, audio, and video model.
- Interleaves audio and video with TMRoPE (Time-aligned Multimodal RoPE).
- Reports state-of-the-art results on Omni-Bench



Takeaway: temporal synchronization is now a first-class multimodal alignment problem.

Sources: Hu et al., MMAAlign, 2025; Qwen2.5-VL, 2025; Qwen2.5-Omni, 2025.

Alignment is now evaluated beyond retrieval

MMEB + VLM2Vec

Massive Multimodal Embedding Benchmark

classification

VQA

retrieval

grounding

36 datasets across 4 meta-tasks.
Instruction-conditioned embeddings
can handle combinations of text and
images instead of only plain caption
matching.
Reported gains are about 10–20% over
prior multimodal embedding baselines.

MMEB-V2 + VLM2Vec-V2

toward images, videos, and visual documents

text

image

video

doc

shared
vector

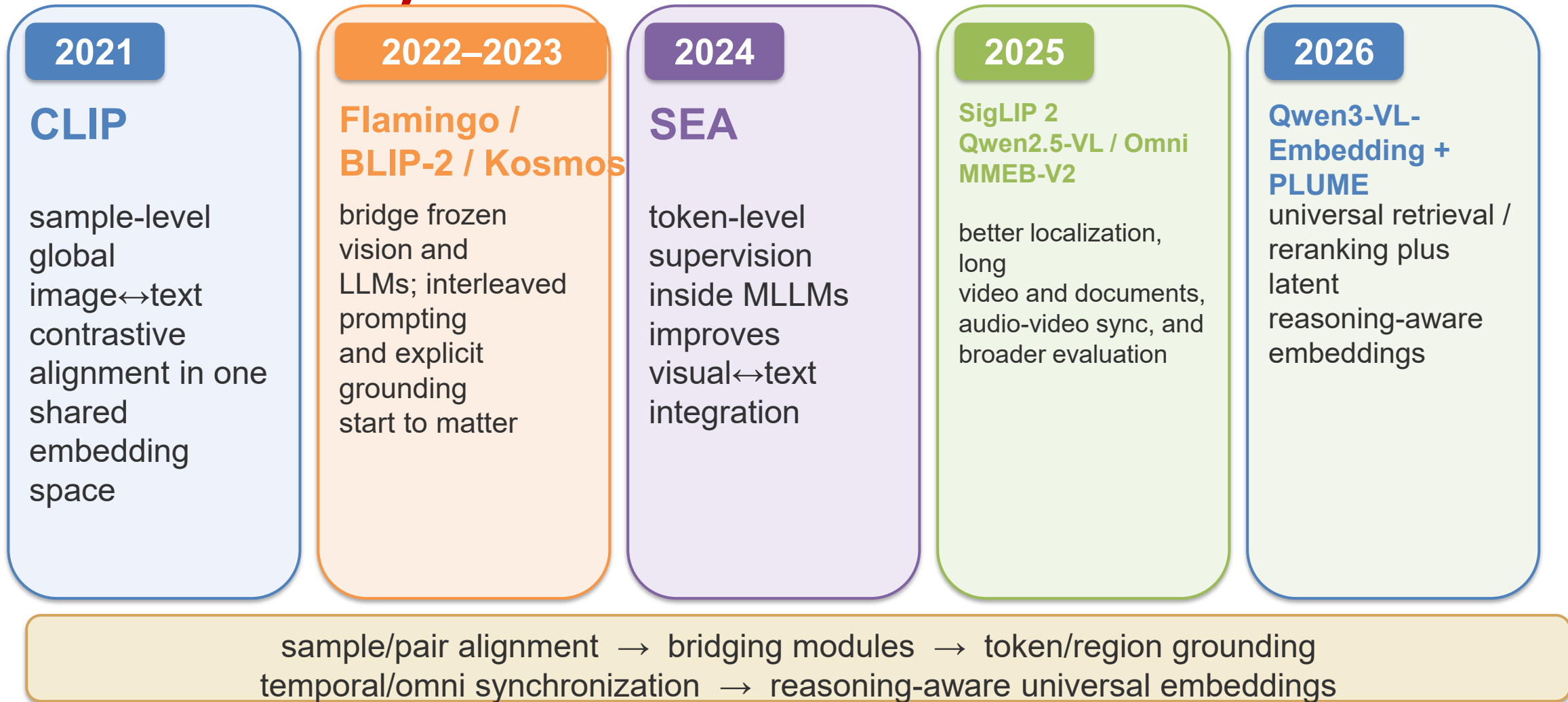
- 78 tasks total.
- video retrieval
- temporal grounding
- video QA / classification
- visual-document retrieval

One framework for text, image, video,
and document inputs.

Benchmark lesson: good alignment must transfer beyond image↔caption retrieval.

Sources: Jiang et al., VLM2Vec / MMEB, 2025; Meng et al., VLM2Vec-V2 / MMEB-V2, 2025.

Timeline: how multimodal alignment evolved (2021→2026)

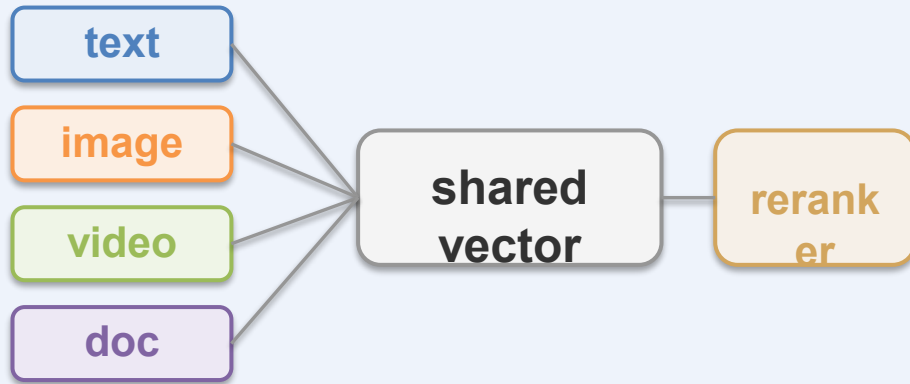


Representative sources: CLIP (2021), Flamingo (2022), BLIP-2 / Kosmos-2 (2023), SEA (2024), SigLIP 2 / Qwen2.5-VL / Omni / MMEB-V2 (2025), Qwen3-VL-Embedding / PLUME (2026).

Early 2026 snapshot: smarter universal multimodal embeddings

Qwen3-VL-Embedding + Reranker

high-precision multimodal search



- unified representation space for text, images, images, and video
- contrastive pretraining → reranker distillation
- supports up to 32k-token inputs; 77.8 on MMEB-V2

PLUME

latent reasoning for universal multimodal embedding



- replaces explicit CoT with a short latent reasoning process
- on MMEB-V2: >30× faster than explicit-CoT baselines
- especially helpful for video and visual-document retrieval

Direction of travel: alignment is becoming retrieval + reranking + efficient intermediate reasoning, not just raw similarity.